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B. Choi, *Introduction to Python Network Automation*

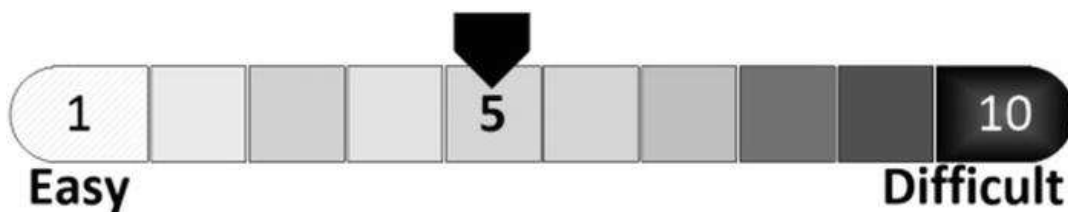
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5. Creating an Ubuntu Server Virtual Machine

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In this chapter, you will be downloading the latest Ubuntu Server 20 bootable image to install and create a virtual machine on your VMware Workstation instance. Ubuntu is a Debian Linux–derivative distro and provides one of the best user experiences on Linux. For most Windows users, Ubuntu Desktop would be the first Linux entry point. However, for this book, we will be using Ubuntu Server 20 instead with a little twist: we will choose to install a UI to add more flexibility to our lab. At the end of this chapter, you will be able to create a Ubuntu Server virtual machine from the ground up, customize selective settings on the operating system, take a snapshot of a virtual machine, and clone a virtual machine



You can create virtual machines (VMs) using bootable image files of different operating systems and load a VM image prepared by others in advance. If you choose the second option and import or convert premade VMs, you will miss out on learning how to build the virtual server from scratch. Anyway, with more powerful desktops and laptops, building and installing a VM from scratch does not take a lot of

time, so you will be guided through the entire installation process starting from downloading the image files to doing basic configurations. Although the VMs in this book are simplified versions of more powerful, scalable enterprise and cloud infrastructure and network solutions, the VM provisioning concepts and techniques are much the same.

In the last five years, the buzzword in IT has been *cloud computing*. In its most basic form, cloud computing is squeezing a lot of powerful servers into centralized or decentralized data centers to provide on-demand services via the Internet. Everything is off-premise, and users are connecting via thin client machines or applications running on remote nodes. Underneath the cloud computing machines are the virtualization technologies, and understanding virtualization means you can get a step closer to cloud services.

After installing VMware Workstation 15 Pro on your laptop, you will need to download a bootable image file for your preferred operating system to build a VM. This image can be any flavor of Windows or Linux image. Our intention is clear that we want to move away from the Windows operating system and get comfortable with the Linux operating system, so our choice is evident. We will download Linux-installable images and start the installation. Since we also want to know the difference between two of the most popular Linux distro versions, namely, Ubuntu and CentOS, we will work with them both. Ubuntu is one of the most popular Debian-based Linux distros, and CentOS is the most widely used Red Hat-based free distribution version. To give you more hands-on experience with both Linux distributions, Ubuntu Server will be installed in this chapter, followed by the CentOS server installation in the next chapter. These two servers will be used throughout this book with full flexibility. The basic kernels are almost identical between the two Linux distributions. Still, there are subtle differences that you can learn if you use them side by side, so installing both and trying to learn basic and general administration tasks will add more value to your learning and to your career.

For the remainder of this chapter, you will concentrate on installing the latest Ubuntu, which will be used as the primary Python network automation server. When you are ready, let's create a Ubuntu Server 20.04 LTS virtual machine .



We are using the latest Ubuntu Server 20.04 LTS and CentOS 8.1 in this book, but in production environments, often system administrators choose to use the $n-1$ version of the latest software. The $n-1$ software concept is to use the previous software version to avoid bugs and software compatibility issues, aiming for the five 9s (99.999 percent uptime each year). Also, you have the flexibility to go one version up if the $n-1$ version does not deliver the performance or services you are looking for.

If you are going to run these servers in the production, consider using Ubuntu 18.04 LTS or CentOS 7.5 for better software compatibility. This makes more sense now for CentOS 8 as it will be supported only until the end of 2021, whereas CentOS 7.5 will still enjoy support until 2024. You'll learn more about this in Chapter 6.

Downloading and Installing a Ubuntu Server 20 Image

To install the latest Ubuntu server , you first need to download a bootable Ubuntu Server image (`.iso`) file from Ubuntu's official download site. Make sure you have a reliable Internet connection for the download and follow the directions in the next sections.



Ubuntu Server 20.04 LTS: what's in a name?

Traditionally, Ubuntu Linux server versions are named with a YY.MM format for the release year and month. So, Ubuntu Server 20.04 indicates that it was first released on the market in the fourth month of

2020. LTS stands for Long Term Support and indicates that this software version is supported for five years as opposed to beta versions. It's the same naming convention for Ubuntu 18.04 LTS, which was released in the fourth month of 2018.

Downloading the Ubuntu Server 20.04 LTS Image

Task

Open your favorite web browser , and then go to the Ubuntu download page. The expected file size is approximately 908MB.

<https://www.ubuntu.com/download/server>

Download the Ubuntu server version, which is the 64-bit PC (AMD64) server install image; the downloaded installable filename is `ubuntu-20.04-live-server-amd64.iso`. See Figure 5-1.

01

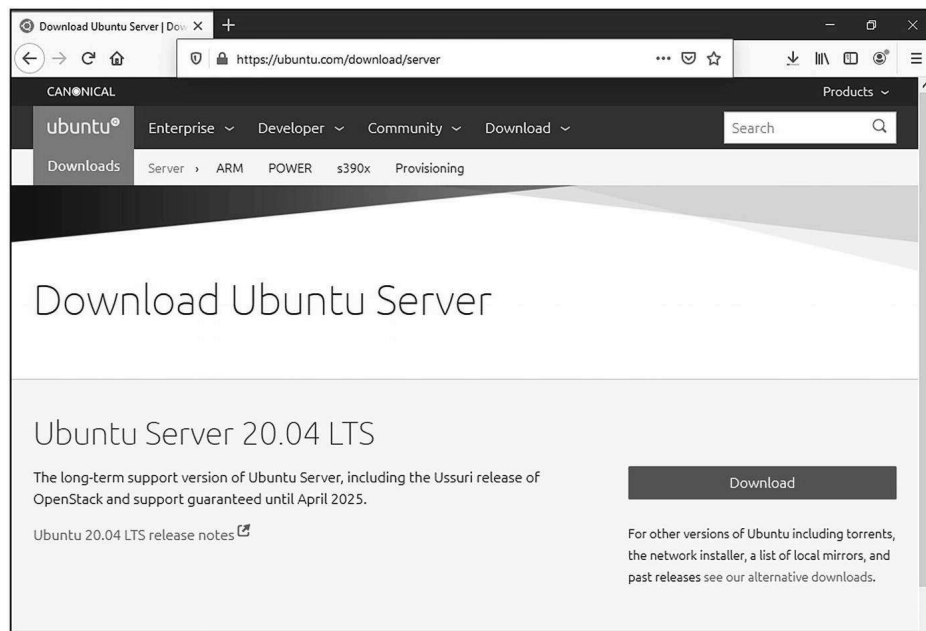


Figure 5-1. Ubuntu Server 20.04 LTS image download page

Task

To check whether the downloaded ISO file has been correctly downloaded, use WinMD5 or a similar tool to check the MD5 value before your installation. See Figure 5-2.

WinMD5 can be downloaded for free from here:

02

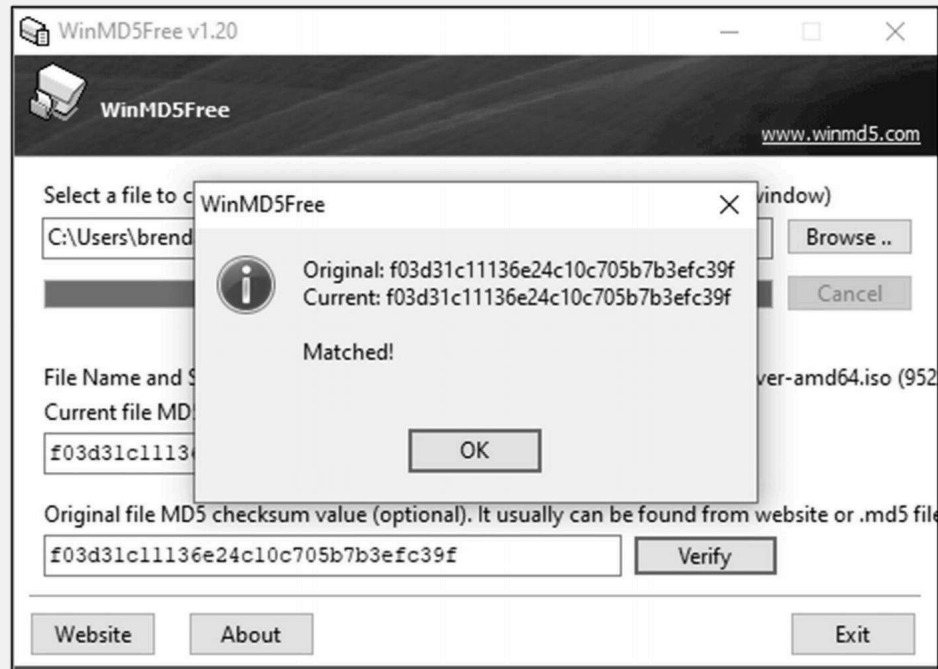


Figure 5-2. Ubuntu Server 20.04 ISO MD5 checksum verification

www.winmd5.com/ or

www.winmd5.com/download/winmd5free.zip

The Ubuntu Server 20 LTS MD5 value can be found at the following link:

<https://releases.ubuntu.com/20.04/MD5SUMS>

You will see something like this:

MD5 value: f03d31c11136e24c10c705b7b3efc39f

Filename: *ubuntu-20.04-live-server-amd64.iso

Installing Ubuntu Server 20.04 LTS

Installing a Linux VM on VMware Workstation is easy, but you have to learn how to create and install various VMs using different bootable ISO images.

The Ubuntu Server 20.04 installation process is as follows:

#	Task
---	------

01	First, start VMware Workstation 15 Pro using the shortcut icon on the host PC (Windows 10). See Figure 5-3 .
----	--



Figure 5-3. VMware Workstation shortcut icon

Task

Then select the File menu and then choose New Virtual Machine, as shown in Figure 5-4.

02

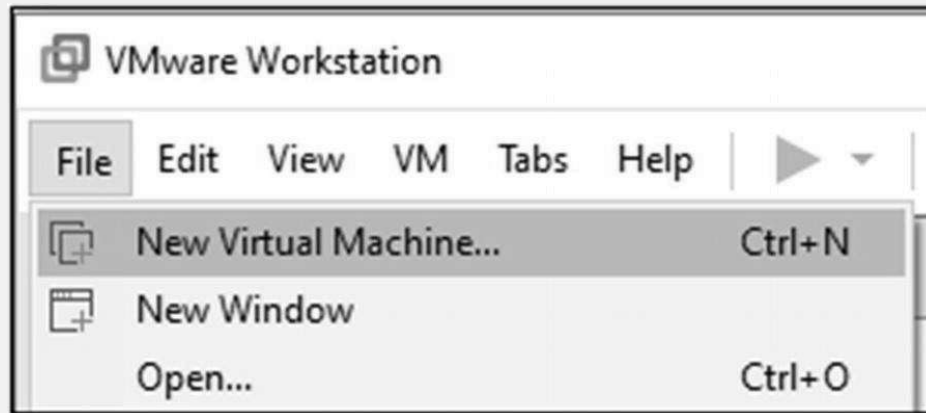


Figure 5-4. Creating a new virtual machine

- 03 When the New Virtual Machine Wizard window appears, leave “Typical (recommended)” selected and click the Next button at the bottom. See Figure 5-5.

Task



Figure 5-5. New Virtual Machine Wizard

- 04 In the next window, select “Installer disk image file (iso)” and click the Browse button on the right. See Figure [5-6](#).

Task

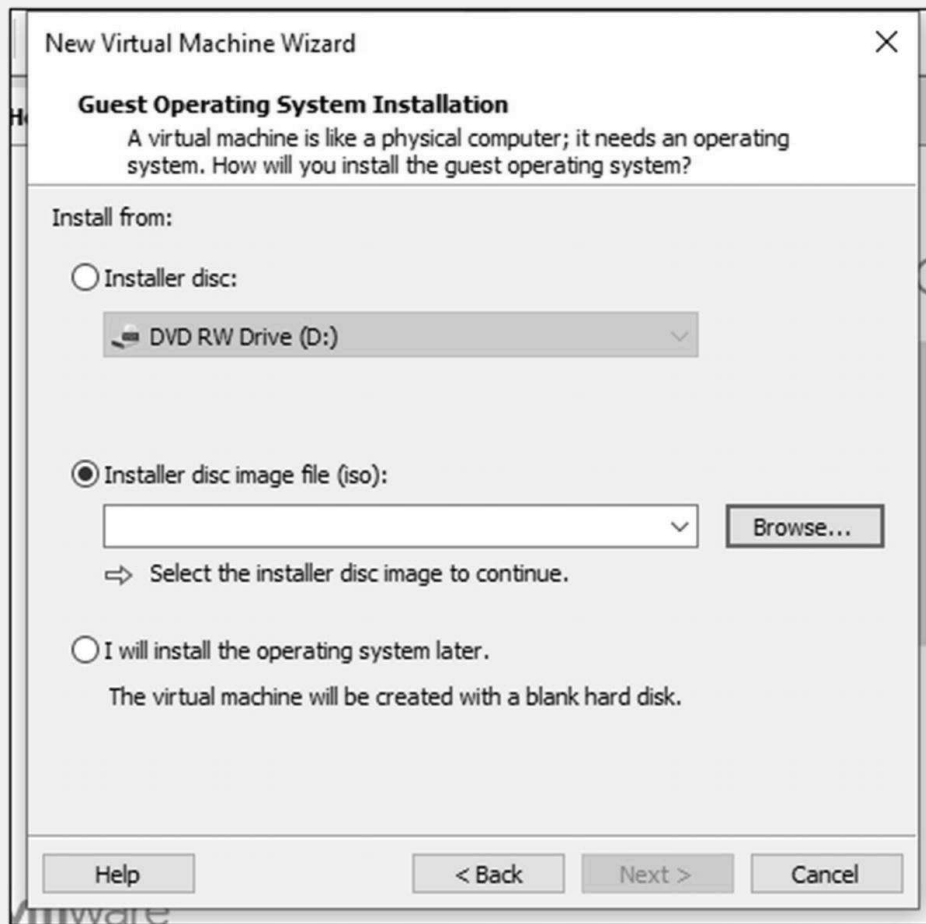


Figure 5-6. Ubuntu installation, selecting installation media

- 05 When the Browse for ISO image window appears, navigate to the Downloads folder where you have downloaded and saved the Ubuntu Server 20.04 installable file and select it. Click the Open button at the bottom. See Figure [5-7](#).

Task

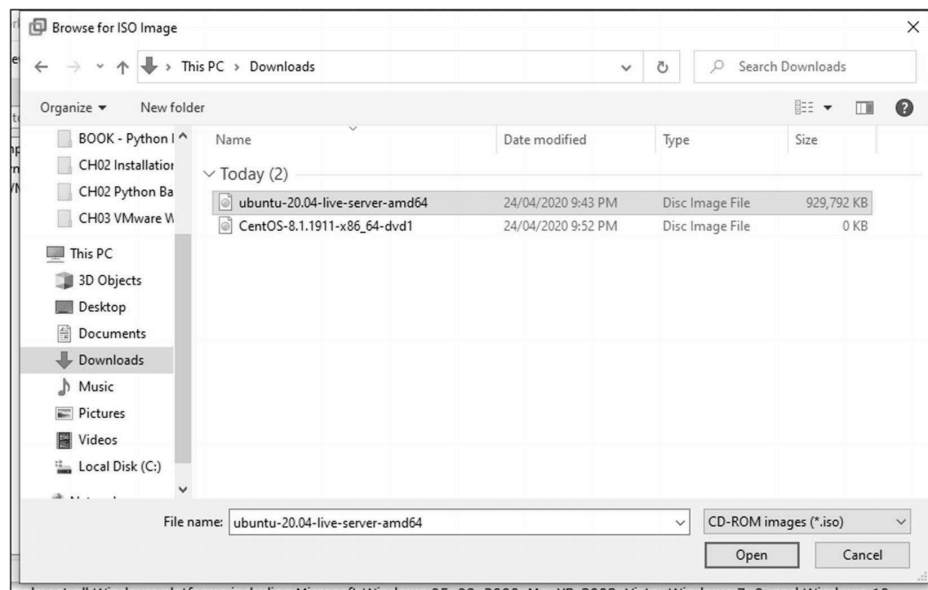


Figure 5-7. Ubuntu installation, selecting an installable image

- 06 When you return to the previous screen, click the Next button. See Figure [5-8](#).

Task

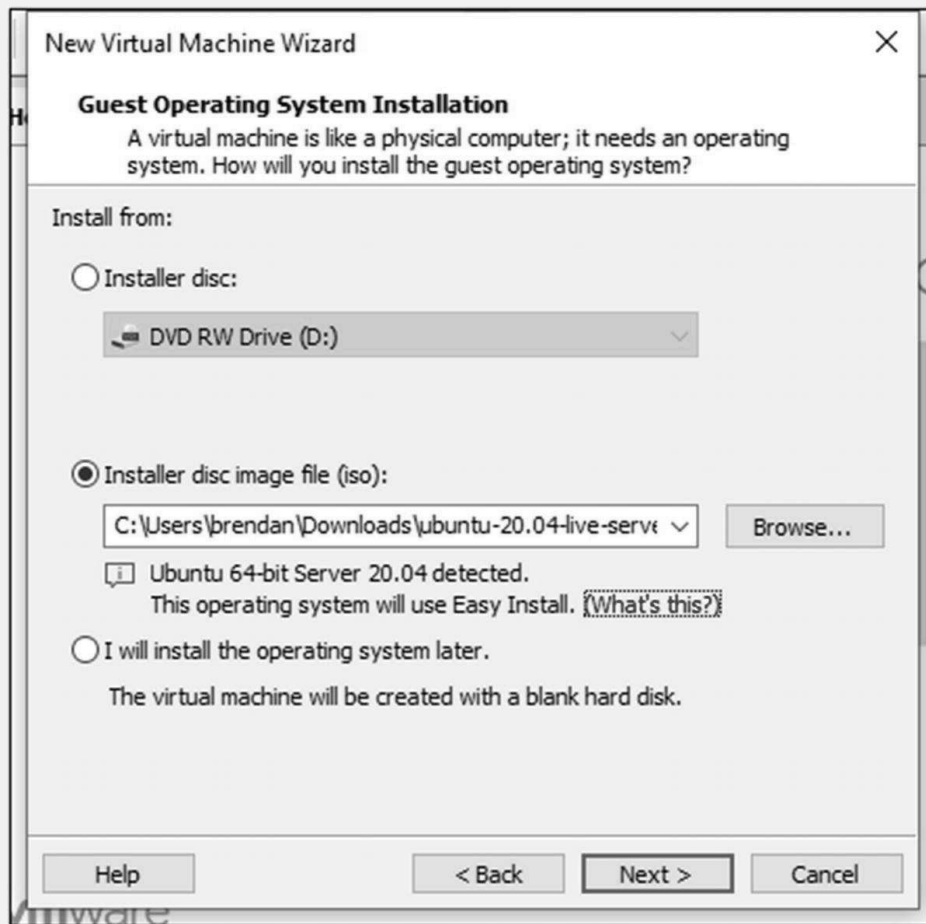


Figure 5-8. Ubuntu installation, selected installable image

- 07 Enter your name, user ID, and password and click the Next button.
See Figure [5-9](#).

Task

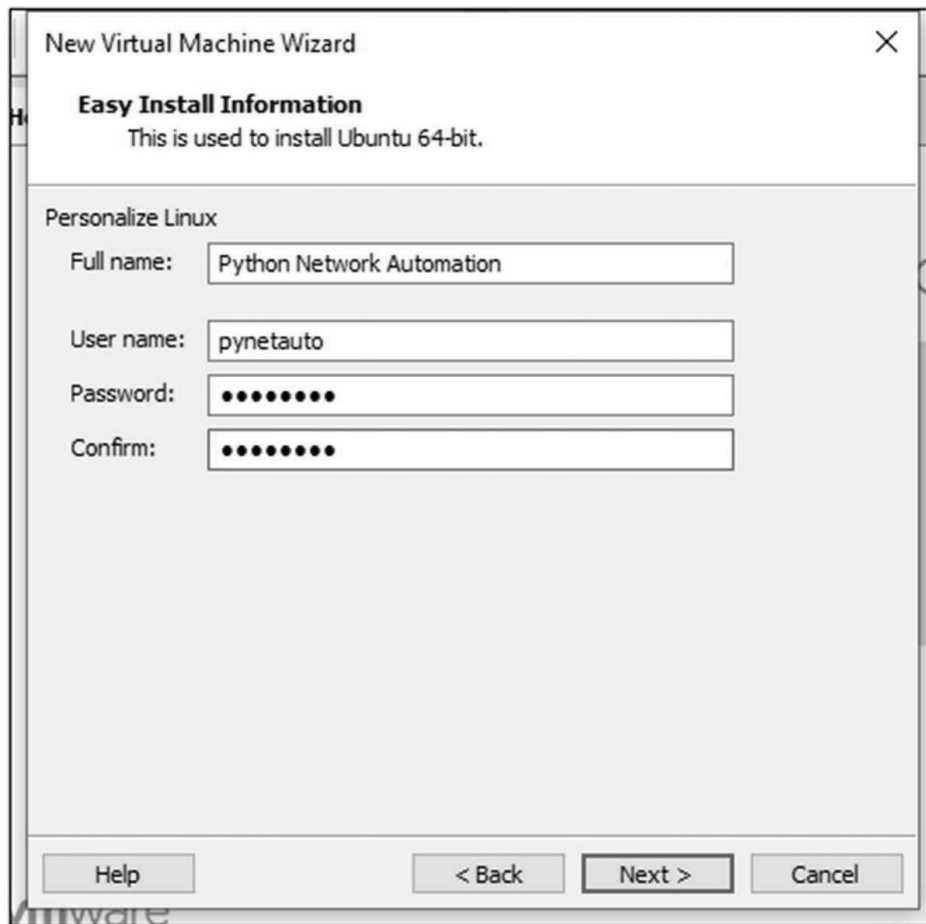


Figure 5-9. Ubuntu installation, entering user credentials

- 08 Choose a meaningful server name for your Ubuntu server. This example is using the default folder, but if you want to store and run your VMs from another folder, you can change the settings now. `ubuntu20s1` is the server name given to my Ubuntu Server 20 server. See Figure [5-10](#).

Task

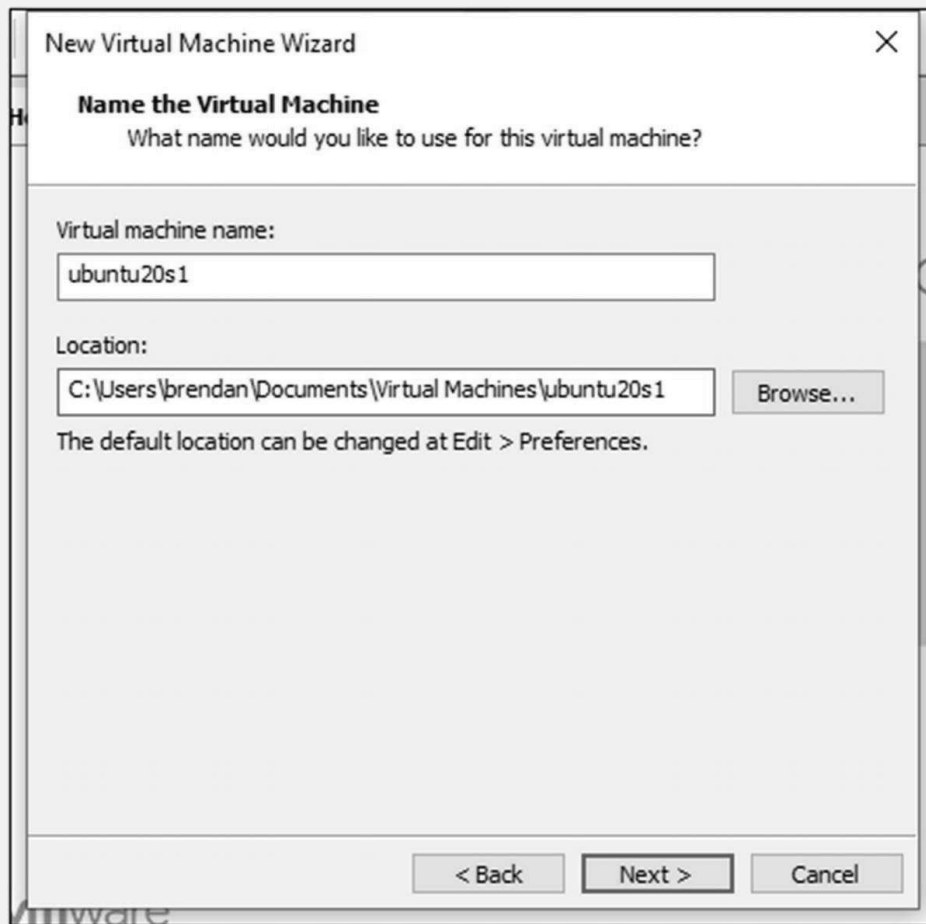


Figure 5-10. Ubuntu installation, VM folder selection

- 09 By default, 20GB will be allocated to your VM. I have enough disk space on my SSD, so I will increase this to 30GB for future use. Also, I prefer to store the virtual disk file as a single file as this makes migration and VM conversions easier. Click the Next button. Since thin provisioning is used, the entire 30GB of disk space is not used, but the disk size will gradually increase on demand while using up the VM's disk space. See Figure [5-11](#).

Task

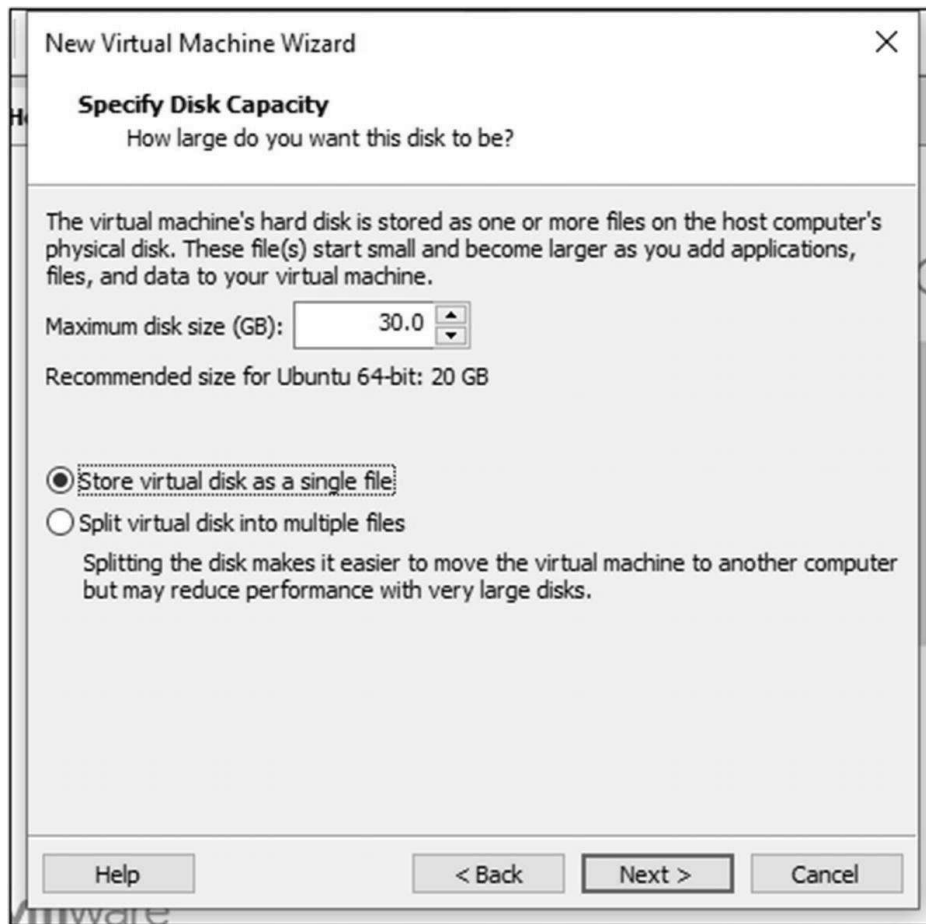


Figure 5-11. Ubuntu installation, disk size allocation, and single virtual disk file option

- 10 On the next screen, click the Finish button with the default setting “Power on this virtual machine after creation” selected. See [Figure 5-12](#).

Task

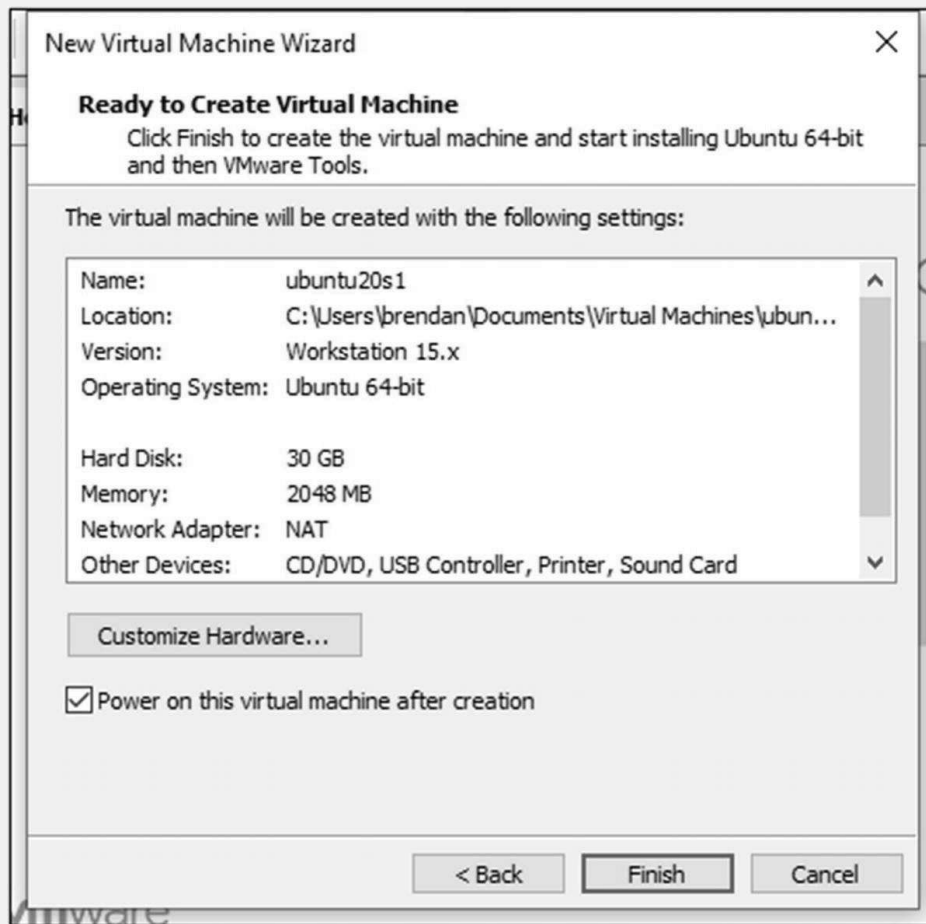


Figure 5-12. Installing Ubuntu, finishing VM configuration

- 11 The new VM will use the Ubuntu boot image to begin the Ubuntu Server installation on the first screen. See Figure [5-13](#).

#

Task

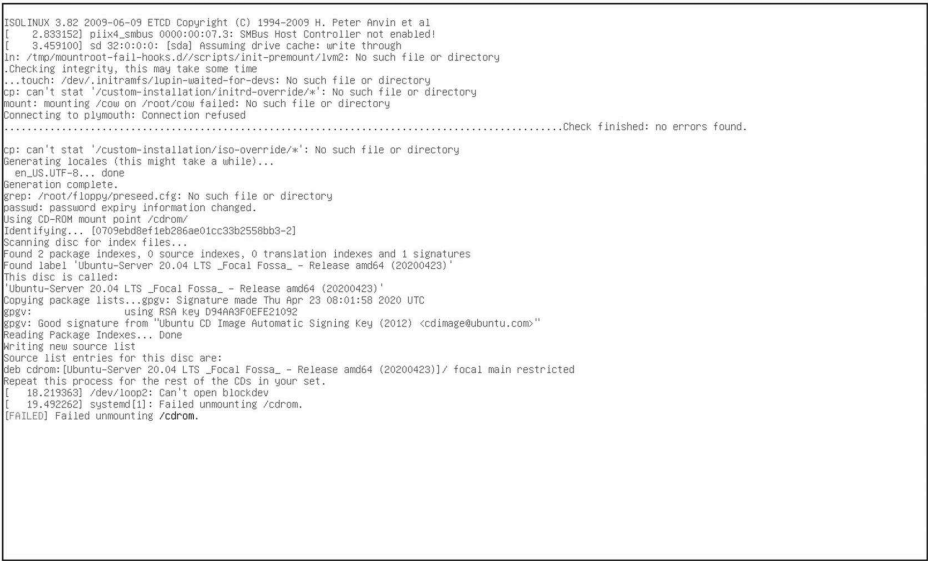


Figure 5-13. Ubuntu installation, server installation’s first screen

Language selection: You are prompted to select a language. English is selected as the default language. If you speak another language, select your native language from the selection. See Figure 5-14.

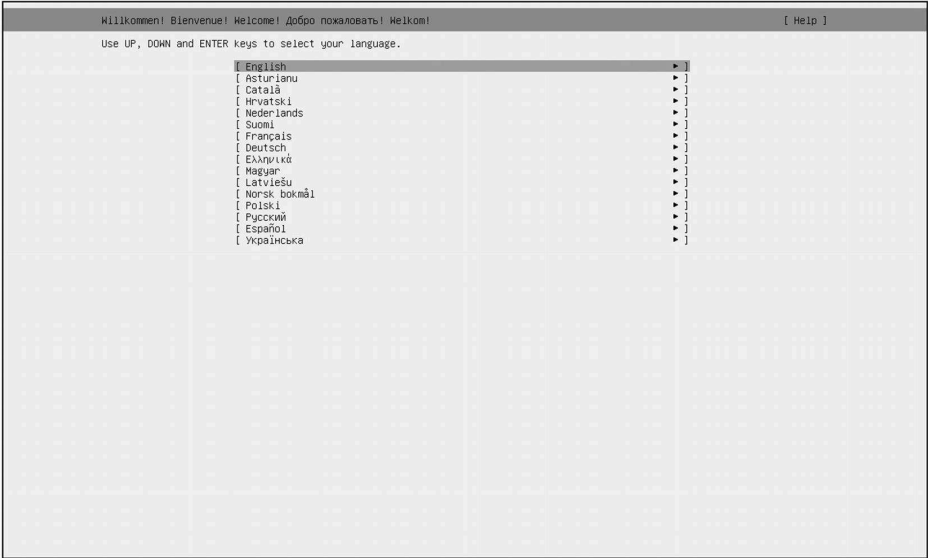


Figure 5-14. Ubuntu installation, default language selection

Task

Keyboard configuration: Leave the keyboard configuration at the default settings, highlight the Done button, and hit the Enter key. See Figure [5-15](#).

13



Figure 5-15. Ubuntu installation, keyboard selection

- 14 *Network connections:* If you did change the default network settings during VM creation, your VM will use a Network Address Translation (NAT) network. The first available IP address will be assigned to your VM from VMware Workstation's NAT DHCP subnet pool, and the default IP range is from 192.168.183.128 to 192.168.183.254. If you changed the subnet, your IP subnet and IP address range will differ from the example shown in this book. If your NAT network's DHCP service has not been disabled, you will receive a valid IP address similar to Figure [5-16](#).

Task

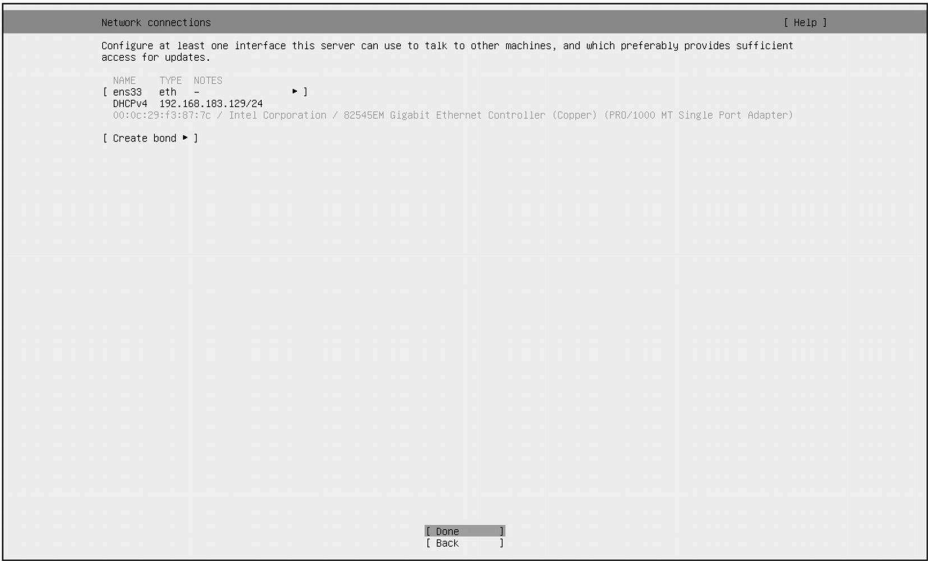


Figure 5-16. Ubuntu installation, network connections

Task

Configure proxy: No proxy server is used on the current network, so there is no need to specify a proxy IP address. However, if you connect to the Internet via your company's proxy server at work, then add the proxy information here. Of course, this information can be changed later. Assuming that you are also connecting via your home network, leave this blank, and let's move on. See Figure **5-17**.

15



Figure 5-17. Ubuntu installation, configuring the proxy

- 16 *Configure Ubuntu archive mirror:* Leave the default selection for this section, then highlight the Done key, and hit the Enter key. See Figure **5-18**.

#

Task



Figure 5-18. Ubuntu installation, configuring the Ubuntu archive mirror

Guided storage configuration: Confirm that the `/dev/sda` size is the same as what you have allocated, then highlight the Done key, and hit the Enter key again. See Figure [5-19](#).



Figure 5-19. Ubuntu installation, guided storage configuration

Task

Storage configuration: Once again, review and confirm the configuration, then highlight Done, and hit the Enter key on your keyboard to move to the next screen. See Figure **5-20**.

18

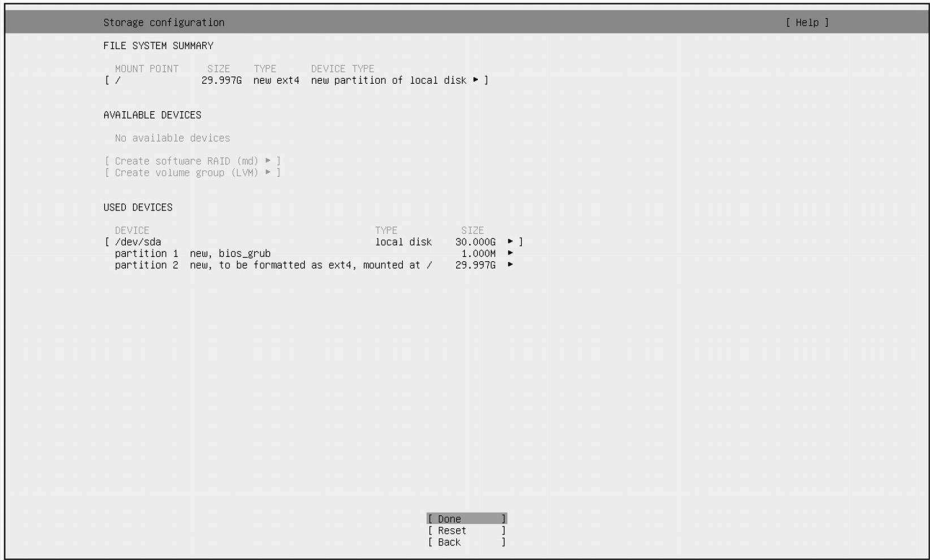


Figure 5-20. Ubuntu installation, storage configuration

19 *Storage configuration:* In the “Confirm destructive action” box, highlight the Continue button and hit the Enter key to complete storage configuration. See Figure **5-21**.

#

Task

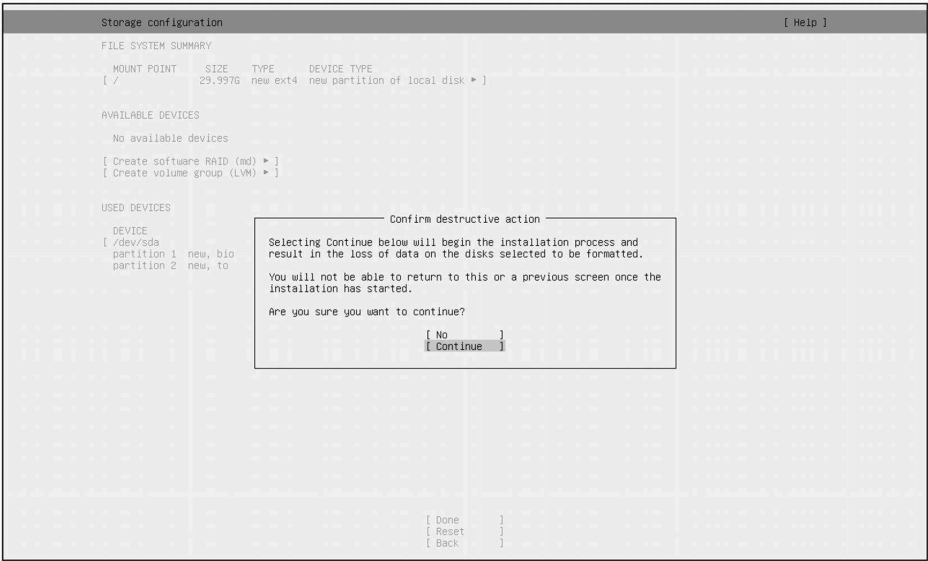


Figure 5-21. Ubuntu installation, storage configuration (confirm destructive action)

Profile setup: Fill in your profile with a unique user ID and password. Make the server’s name the same as what you have specified during VM creation; in this case, it is ubuntu20s1 for Ubuntu Server 20 1. Highlight the Done key and hit the Enter key again. See Figure 5-22.



Figure 5-22. Ubuntu installation, setting up the profile

Task

SSH Setup: You will connect to this server via an SSH connection, so installing the OpenSSH server for an SSH connection is recommended. We can install this service after the OS installation, but it makes sense to install the SSH server now rather than later. Use the spacebar to select the “Install OpenSSH server” option, then highlight the Done key, and hit the Enter key again. See [Figure 5-23](#).

21



Figure 5-23. Ubuntu installation, setting up SSH

- 22 *Featured Server Snaps:* If you plan to use any of the services immediately after the installation, make your selections using the spacebar. Docker and PowerShell have been selected in [Figure 5-24](#), as they are often useful services on any Linux server. Once all the selections are made, then highlight the Done key and hit the Enter key on your keyboard.

Task

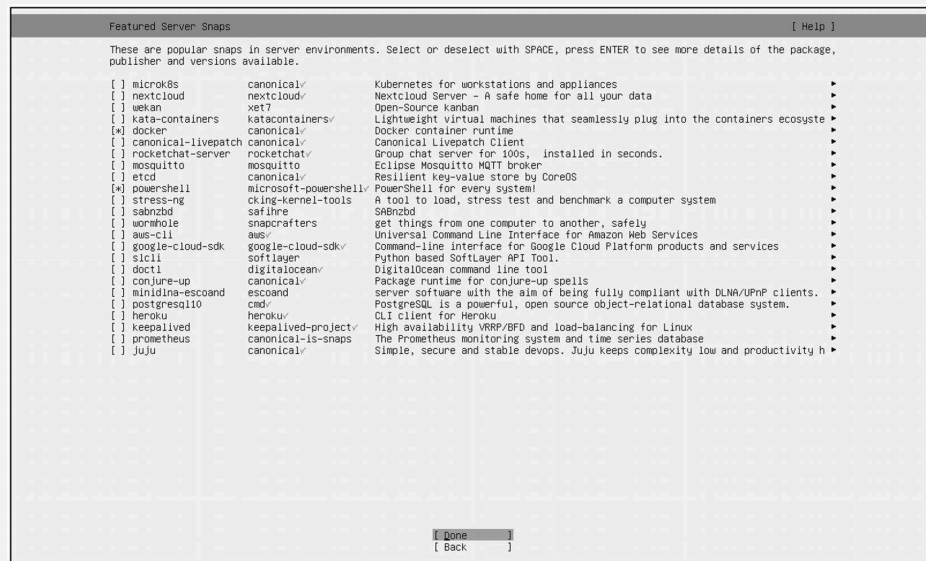


Figure 5-24. Ubuntu installation, featured server snaps

Installation complete: Once you get to the final installation screen, click the Reboot option, and press the Enter key to restart your first VM. See Figure [5-25](#).

23



Figure 5-25. Ubuntu installation, installation completed

24 After the reboot, your screen will look similar to Figure [5-26](#).

Task

```
Ubuntu 20.04 LTS ubuntu20s1 tty1
ubuntu20s1 login:          Mounting Mount unit for core18, revision 1705...
[ OK ] Mounted Mount unit for core18, revision 1705.
      Mounting Mount unit for lxd, revision 14804...
[ OK ] Mounted Mount unit for lxd, revision 14804.
[ OK ] Listening on Socket unix for snap application lxd.daemon.
      Starting Service for snap application lxd.activate...
[ OK ] Finished Service for snap application lxd.activate.
[ OK ] Finished Wait until snapd is fully seeded.
      Starting Apply the settings specified in cloud-config...
[ OK ] Reached target Multi-User System.
[ OK ] Reached target Graphical Interface.
      Starting Update UTMP about System Runlevel Changes...
[ OK ] Finished Update UTMP about System Runlevel Changes.
      Mounting Mount unit for core, revision 8935...
[ OK ] Mounted Mount unit for core, revision 8935.
```

Figure 5-26. Ubuntu operation rebooted, initial screen

- 25 Now use your user ID and password to log into your Ubuntu VM. Now your server is ready to serve you, but how well you will use this server in your study is entirely up to you. See Figure [5-27](#).

Task

```

ubuntu20s1 login: pynetauto
Password:
Welcome to Ubuntu 20.04 LTS (GNU/Linux 5.4.0-26-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Fri Apr 24 13:53:13 UTC 2020

System load: 0.06          Processes:            191
Usage of /:  14.4% of 29.40GB Users logged in:        0
Memory usage: 17%          IPv4 address for ens33: 192.168.183.129
Swap usage:  0%

0 updates can be installed immediately.
0 of these updates are security updates.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

[ OK ] Created slice User Slice of UID 1000.
        Starting User Runtime Directory /run/user/1000...
[ OK ] Finished User Runtime Directory /run/user/1000.
        Starting User Manager for UID 1000...
[ OK ] Started User Manager for UID 1000.
[ OK ] Started Session 1 of user pynetauto.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

pynetauto@ubuntu20s1:~$

```

Figure 5-27. Ubuntu installation, first login screen

Logging In to a New Ubuntu Server 20 via SSH

As you will soon discover, accessing the new VM via VMware Workstation's console screen is cumbersome. An easier way to manage our server is to connect to this server via an SSH connection, using an SSH client software from our Windows host PC. This is like how system administrators manage their systems on their network. Suppose you have already memorized the server's allocated IP address during the Ubuntu server installation process in the "Network connections" section. In that case, you have a sharp mind, but if you have forgotten the assigned IP address, let's find the IP address so we can SSH into the server.

- 01a The first and most straightforward way to find the IP address on an Ubuntu is to issue an IP address or `IP add`s command. A NATted IP address from the VM NAT network should be assigned to your ens33 network adapter. The server in this example received the 192.168.183.129/24 IP address. Your server's allocated IP address does not have to match, but if you have a valid IP address, that is a good indicator that your VM's network

is working correctly. If you received an IP address from a different subnet or have not received a valid IP address, then you must go back and review your VM network editor or VM network settings. See Figure [5-28](#).

```
pynetauto@ubuntu20s1:~$ ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:f3:87:7c brd ff:ff:ff:ff:ff:ff
    inet 192.168.183.129/24 brd 192.168.183.255 scope global dynamic ens33
        valid_lft 1134sec preferred_lft 1134sec
    inet6 fe80::20c:29ff:fef3:877c/64 scope link
        valid_lft forever preferred_lft forever
```

Figure 5-28. Ubuntu operation, finding the server IP address using ip add

- 01b The second and more traditional way to find the IP address on any Linux server is to issue the `ifconfig` command. If you are only starting to transition to Linux OS from Windows OS, you will immediately notice that this command is like an `ipconfig` command at the Windows command prompt. On Ubuntu Server 20.04, we need to install `net-tools` first to enable this command. Run the `sudo apt install net-tools` command to install `net-tools`; this is also an excellent test for your Internet connectivity from the virtual Ubuntu server. See Figure [5-29](#).

```
pynetauto@ubuntu20s1:~$ ifconfig

Command 'ifconfig' not found, but can be installed with:

sudo apt install net-tools

pynetauto@ubuntu20s1:~$ sudo apt install net-tools
[sudo] password for pynetauto: _
```

Figure 5-29. Ubuntu operation, net-tools installation to enable the ifconfig command

Once the installation has been completed, issue the `ifconfig` command to check the IP address. Because we installed the Docker service during the Ubuntu installation, the tool even picked up the `docker0` interface and is displaying it in Figure [5-](#)

30. But here, we are only interested in the information from the ens33 network adapter, and as expected, an IP address of 192.168.183.129/24 is given to the ens33 adapter with a broadcast address of 192.168.183.255.

```
pynetauto@ubuntu20s1:~$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 02:42:be:ac:2f:e3 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.183.129 netmask 255.255.255.0 broadcast 192.168.183.255
    inet6 fe80::20c:29ff:fef3:877c prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:f3:87:7c txqueuelen 1000 (Ethernet)
    RX packets 86901 bytes 131309226 (131.3 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 27333 bytes 1657185 (1.6 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 106 bytes 8382 (8.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 106 bytes 8382 (8.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Figure 5-30. Ubuntu operation, finding the server IP address using ifconfig

02 Stop here for one moment and think! The gateway IP for the 192.168.183.0/24 subnet is 192.168.183.2, not 192.168.183.1. 192.168.183.2 is the NAT gateway IP, and 192.168.183.1 is the assigned IP address of the Windows 10 host's VMnet8 (VMware Network Adapter VMnet8) interface. See Figure **5-31**.

```
pynetauto@ubuntu20s1:~$ ip route | grep ^default
default via 192.168.183.2 dev ens33 proto dhcp src 192.168.183.129 metric 100
```

Figure 5-31. Ubuntu operation, confirming default gateway on Linux

You can confirm this information from the Ubuntu server console by running `ip route | grep ^default` (or `ip r | grep ^def`). The meaning of this command is to look at the IP route with a general regular expression (`grep`) and find a string that begins with `^` and the word `default`. See Figure **5-32**.

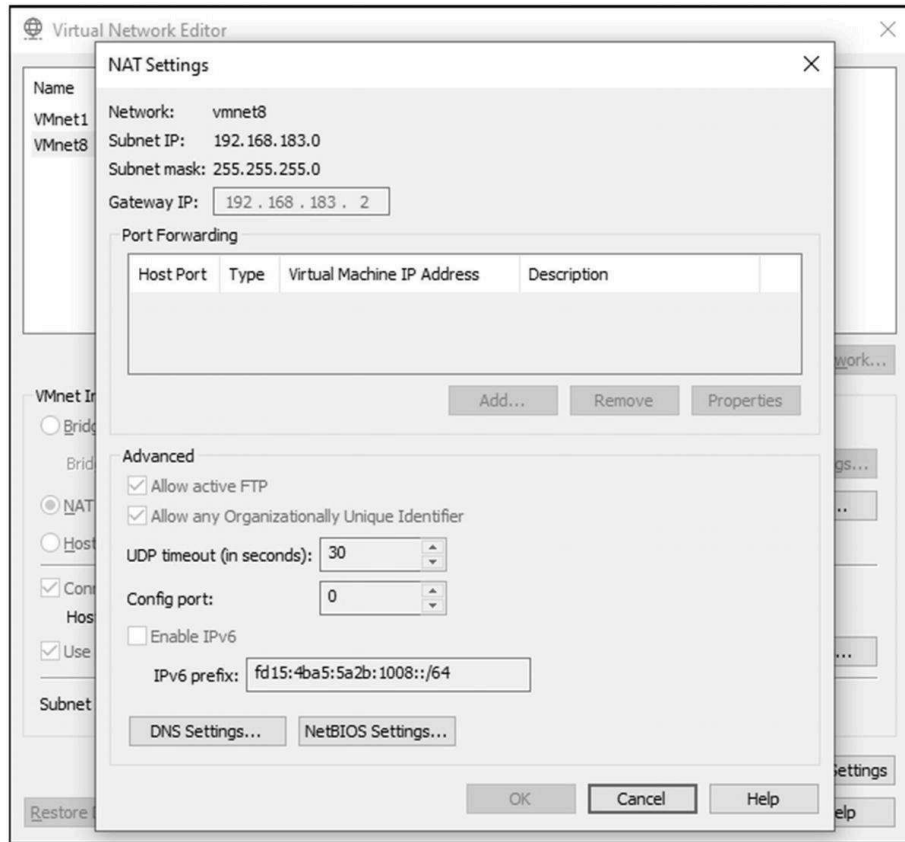


Figure 5-32. Ubuntu operation , confirming default gateway on VMware Workstation

Run the `netsh interface ip show addresses "VMware Network Adapter VMnet8"` command from your Windows 10 host PC. See Figure [5-33](#).

```
C:\Users\brendan>netsh interface ip show addresses "VMware Network Adapter VMnet8"

Configuration for interface "VMware Network Adapter VMnet8"
DHCP enabled:                Yes
IP Address:                  192.168.183.1
Subnet Prefix:               192.168.183.0/24 (mask 255.255.255.0)
InterfaceMetric:             35
```

Figure 5-33. Ubuntu operation, VMnet8 host IP address

03 Now, if you have not yet downloaded PuTTY from the Internet, go to PuTTY's home page (<https://www.putty.org/>) and download a copy of PuTTY. Launch PuTTY from your Windows 10 host and then type in Ubuntu Server's IP address with port 22 for

the SSH connection. Click the Open button at the bottom-right corner of the PuTTY Configuration window. See Figure 5-34.

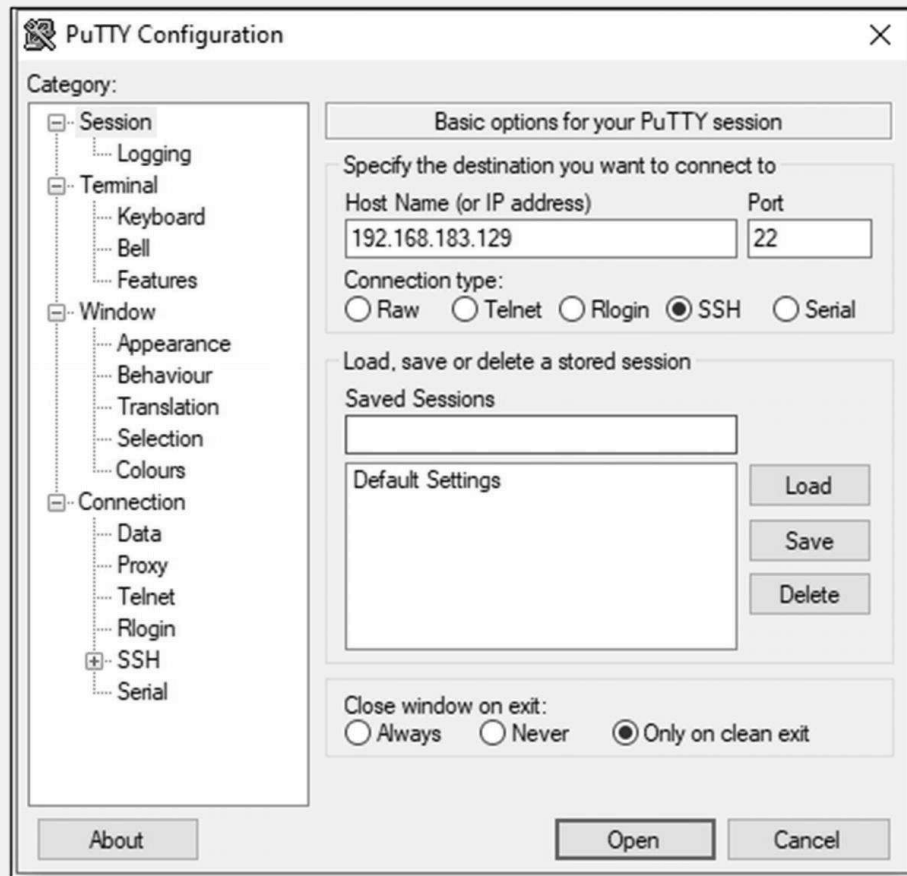


Figure 5-34. Ubuntu operation, SSH client login from PuTTY

- 04 The server will send the server's rsa2 key fingerprint to your PuTTY (SSH) session. Click Yes to proceed to the server's login screen. See Figure 5-35.



Figure 5-35. Ubuntu operation, accepting the PuTTY security alert from the server

Enter your username and password and hit the Enter key on your keyboard to log into your Ubuntu server. Now you are connected to your server via SSH. See Figure [5-36](#).

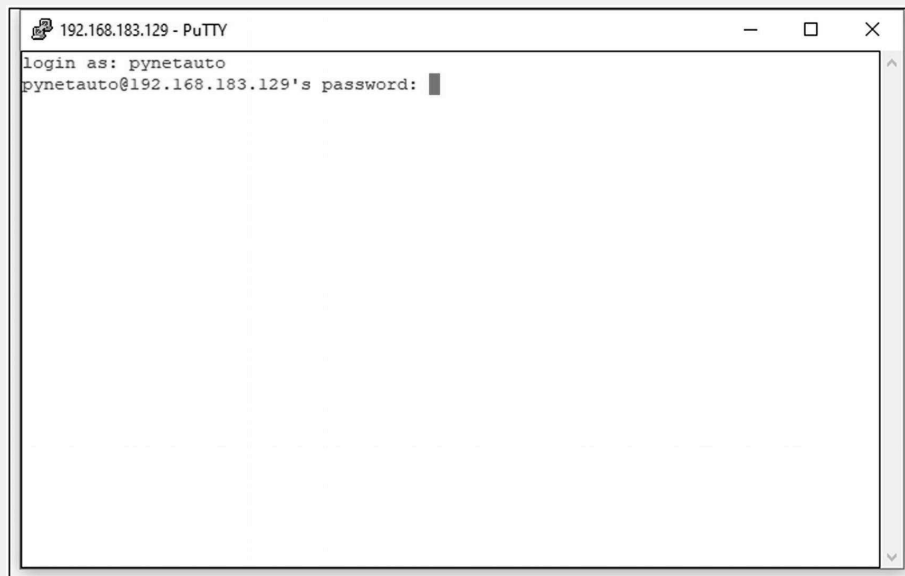
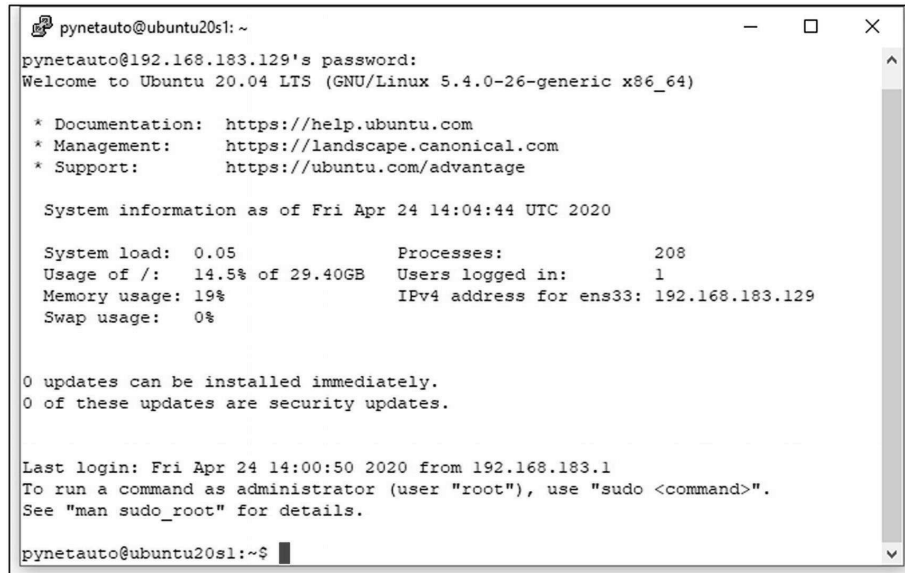


Figure 5-36. Ubuntu operation, SSH server login

- 05
- 06 Customize the PuTTY configuration to suit your preferences. In Figure [5-37](#), I have changed the font and background color of my PuTTY console session and then saved the settings.



```
pynetauto@ubuntu20s1: ~  
pynetauto@192.168.183.129's password:  
Welcome to Ubuntu 20.04 LTS (GNU/Linux 5.4.0-26-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/advantage  
  
System information as of Fri Apr 24 14:04:44 UTC 2020  
  
System load:  0.05          Processes:            208  
Usage of /:   14.5% of 29.40GB Users logged in:       1  
Memory usage: 19%          IPv4 address for ens33: 192.168.183.129  
Swap usage:   0%  
  
0 updates can be installed immediately.  
0 of these updates are security updates.  
  
Last login: Fri Apr 24 14:00:50 2020 from 192.168.183.1  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
pynetauto@ubuntu20s1:~$
```

Figure 5-37. Ubuntu operation, PuTTY background and font color change

Customize Ubuntu Server

After creating your Ubuntu Server 20 LTS virtual machine, it works out of the box; however, Ubuntu has a few critical settings disabled by default. We want to get full access to these settings for our testing lab purposes to add more flexibility. We can tweak a few things to make this server more user-friendly to add more value to your learning. If you are running an Ubuntu server in production, do not enable these settings as they are not the best practice. Since this is our lab VM, we are going to perform some system customizations. First, enable `root` user `ssh` login from remote clients; this allows you to connect to this server via SSH as the root user, saving time during system administration. Second, install a desktop GUI; this will allow you to log into the desktop settings and then use it as a standard end-user machine and use the available applications via the Linux GUI. Third, enable root user GUI access. This feature is disabled by default too, but root user GUI access to the server may be required for quick troubleshooting.

Ubuntu VM Customization 1: Enable Root User SSH Login on Ubuntu Server 20.04

In general, most recent Ubuntu servers do not allow root users to log into the terminal console or log in via SSH for security reasons. Still, this security feature can often slow you down during your lab scenarios, demanding you to enter `sudo` in front of every command you type in and deny you permission to work with flexibility in your lab. For this reason, we can allow SSH logins to add a little flexibility to your lab. However, it is best practice to leave the root console and SSH login disabled for security purposes in the real production environment. First, to enable direct root user SSH access, log into your server as yourself, and run the command `sudo passwd`. Type in your password, and you are immediately prompted to enter a new UNIX password. Entering the password twice here will enable the root user password. Asterisks are used for illustration purposes, but the password will be hidden while you are typing it.

Task

Enable the root user password by performing the following tasks:

```
pynetauto@ubuntu20s1:~$ sudo passwd
01 sudo password for pynetauto: *****
Enter new UNIX password:*****
Retype new UNIX password:*****
passwd: password updated successfully
```

02 The previous action does not automatically allow SSH login to the server; this needs to be manually enabled by updating the SSH server configuration file. To enable root user SSH login, take the next steps:

```
pynetauto@ubuntu20s1:~$ sudo nano
/etc/ssh/sshd_config
```

■ **Note** If you have logged in as a root user at this stage, you do not have to add `sudo` in front of your command.

If you have been using the older Ubuntu version, first you may have to install the nano text editor or use vi as your text

Task
editor.

The line starting with `#PermitRootLogin prohibit-password` prevents root user SSH login, and we have to uncomment this line to allow root user SSH login. For simplicity and best practices, leave the default line unchanged and add a new line as you may have to restore the configuration to the original setting. Add the following line after the original line and save the file:

`PermitRootLogin yes`

See Figure [5-38](#).

03

```

pynetauto@ubuntu20s1: ~
GNU nano 4.8 /etc/ssh/sshd_config
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
^X Exit ^R Read File ^\ Replace ^U Paste Text ^T To Spell ^_ Go To Line

```

Figure 5-38. Ubuntu VM customization 1, changing sshd_config

04

To make the previous change take effect, you must restart the SSH server service.

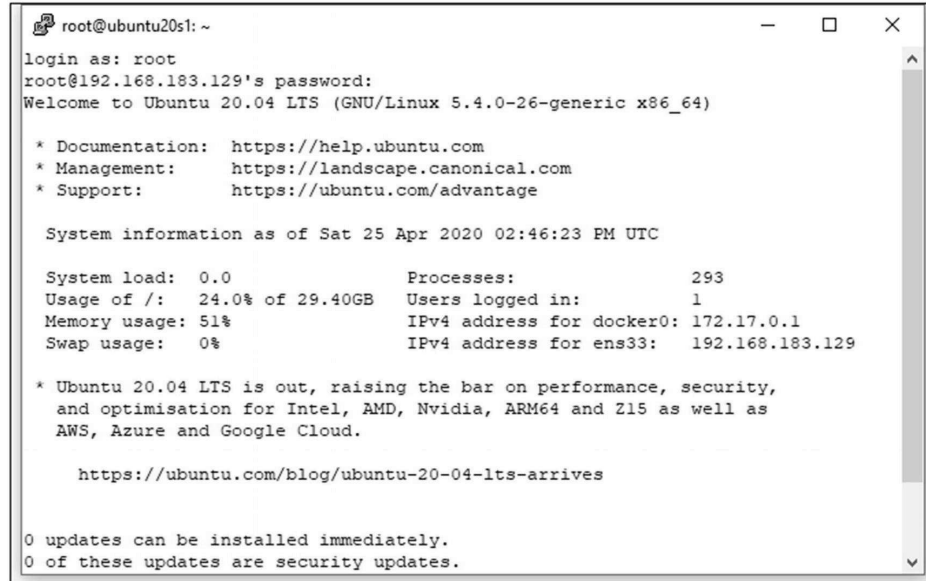
`pynetauto@ubuntu20s1:~$ sudo service ssh restart`

05

Now from your Windows 10 host PC, launch PuTTY, enter the IP address of your server, and SSH in with port 22. Once you

Task

successfully logged into your server via an SSH session, you will see a screen similar to Figure 5-39.



```

root@ubuntu20s1: ~
login as: root
root@192.168.183.129's password:
Welcome to Ubuntu 20.04 LTS (GNU/Linux 5.4.0-26-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Sat 25 Apr 2020 02:46:23 PM UTC

System load:  0.0               Processes:           293
Usage of /:   24.0% of 29.40GB   Users logged in:    1
Memory usage: 51%              IPv4 address for docker0: 172.17.0.1
Swap usage:   0%                IPv4 address for ens33:  192.168.183.129

 * Ubuntu 20.04 LTS is out, raising the bar on performance, security,
   and optimisation for Intel, AMD, Nvidia, ARM64 and Z15 as well as
   AWS, Azure and Google Cloud.

https://ubuntu.com/blog/ubuntu-20-04-lts-arrives

0 updates can be installed immediately.
0 of these updates are security updates.

```

Figure 5-39. Ubuntu VM customization 1, root user SSH login



You can use a single-line command to grant the SSH access to the root user. Simply run the command from your terminal session.

```
sudo sed -i 's/#PermitRootLogin prohibit-
password/PermitRootLogin yes/' /etc/ssh/sshd_config
```

```
[sudo] password for pynetauto: *****
```

Ubuntu VM Customization 2: Install a Desktop GUI and Other Packages

Optionally, you can install a desktop GUI to run on an Ubuntu Server. In a testing lab, GUI can provide the extra features required while testing various IT networking scenarios, for example testing an HTTPS connection via a web browser. In a typical production environment, Linux is usually installed as a

headless server without a GUI for increased server performance and security; a GUI for Linux server is not a prerequisite in the production environment. Anyway, you should know how to install a GUI on Ubuntu Server, so let's learn how to install a desktop GUI and other software packages from the initial Ubuntu Server 20 terminal console.

Task

First, we have to install `tasksel` on the newly installed Ubuntu Server 20.04. Run the following command from the VMware Workstation Console or SSH session:

```
sudo apt install tasksel
```

See Figure [5-40](#).

01

```
pynetauto@ubuntu20s1:~$ tasksel
Command 'tasksel' not found, but can be installed with:

sudo apt install tasksel

pynetauto@ubuntu20s1:~$ sudo apt install tasksel
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  laptop-detect tasksel-data
The following NEW packages will be installed:
  laptop-detect tasksel tasksel-data
0 upgraded, 3 newly installed, 0 to remove and 2 not upgraded.
Need to get 40.0 kB of archives.
After this operation, 309 kB of additional disk space will be used.
Do you want to continue? [Y/n] _
```

Figure 5-40. Ubuntu VM customization 2, installing tasksel

- 02 After the `tasksel` installation finishes, you can run `tasksel` by issuing the `sudo tasksel` command. This will launch the Ubuntu package configuration application to select and install other useful applications; Ubuntu desktop is one option under this package manager.
- Select “Ubuntu desktop” and move to OK to start the desktop installation. See Figure [5-41](#).

Task

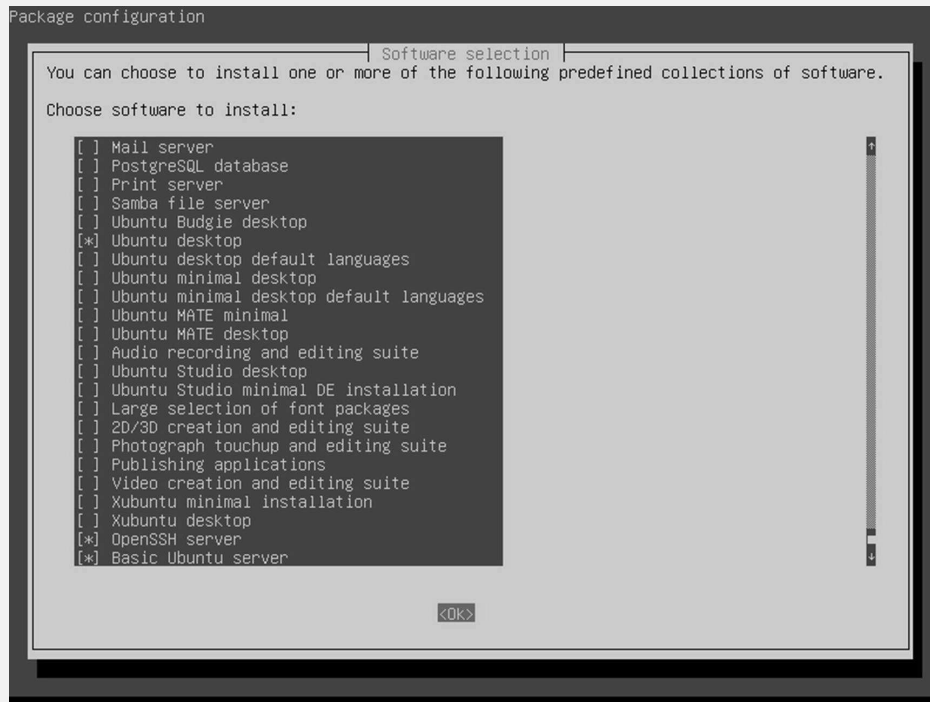


Figure 5-41. Ubuntu VM customization 2, selecting “Ubuntu desktop” (GUI)

- 03 Wait until the desktop installation is 100 percent complete, and then issue the `sudo reboot` command to restart your server. See Figure [5-42](#).

Task

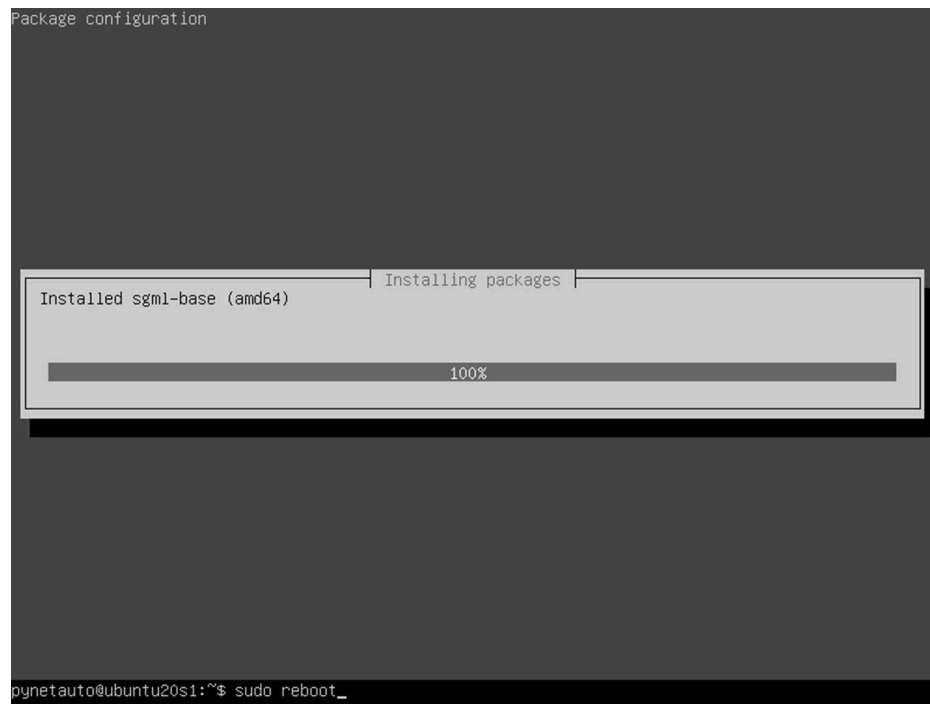


Figure 5-42. Ubuntu VM customization 2, Ubuntu desktop installation

Task

After the reboot , you will be prompted with the Ubuntu GUI login interface. Enter your login credentials and log into your server. See Figure **5-43**.

04

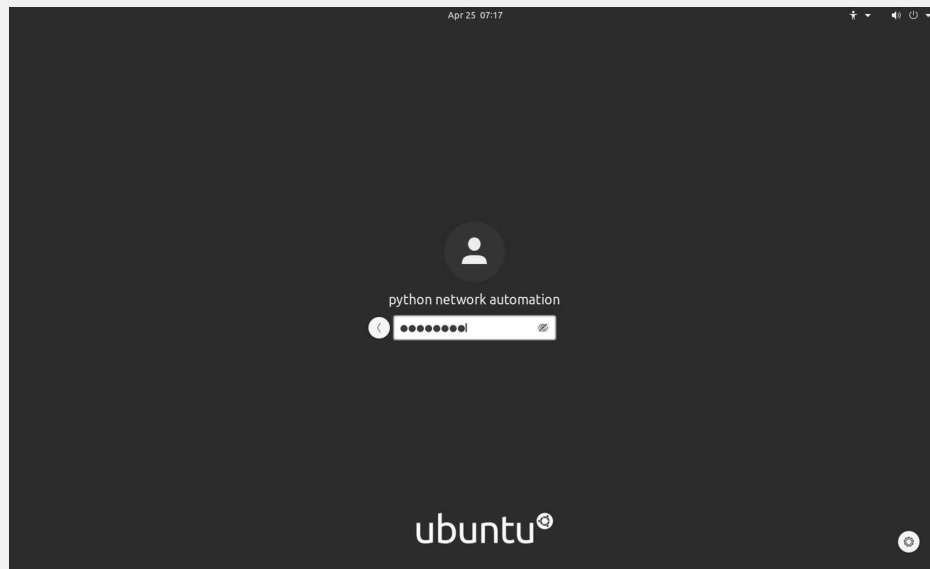


Figure 5-43. Ubuntu VM customization 2 , Ubuntu desktop login

05 You have successfully installed Ubuntu desktop on an Ubuntu Server. See Figure **5-44**.

Task

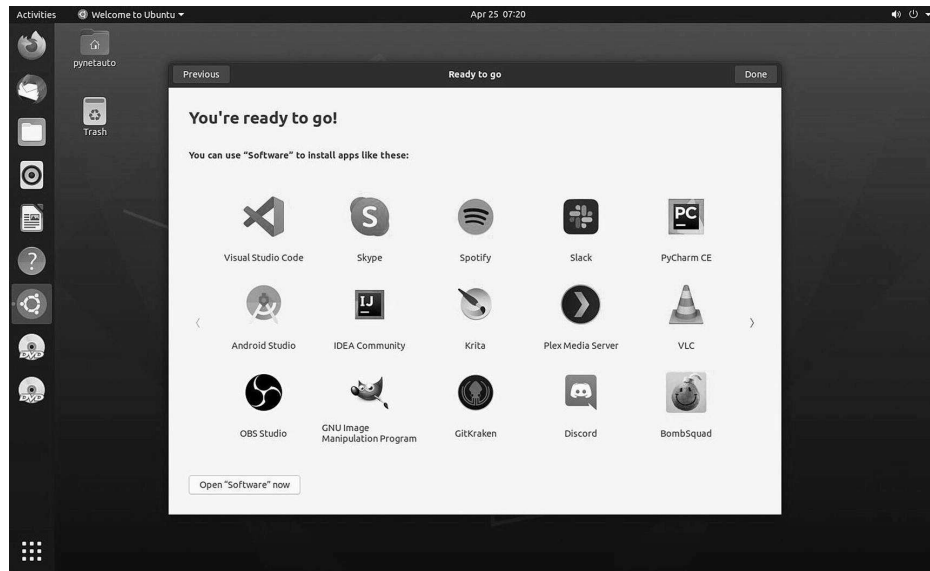


Figure 5-44. Ubuntu VM customization 2, Ubuntu desktop GUI

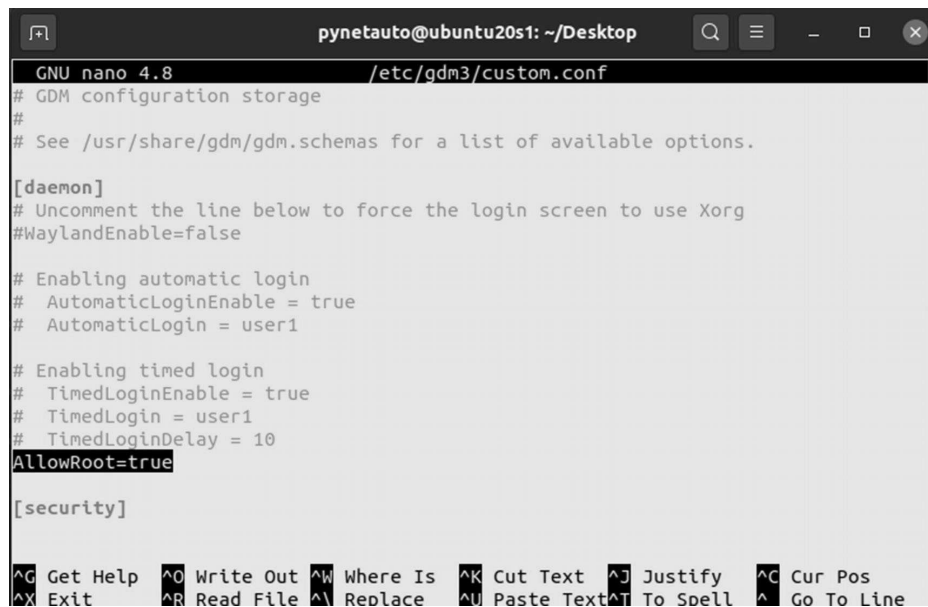
Ubuntu VM Customization 3: Enable Root User GUI Access

If you have sharp eyes, you probably noticed that root user GUI login does not work, and you are questioning , how can I enable root user GUI login? Since this is our lab server and we can explore and customize as much as we want, let's enable the root user GUI login. First, you have to be logged into the server as a normal user (yourself) to complete this task or, optionally, SSH into the server from PuTTY.

Task

- 01 Open and edit `/etc/gdm3/custom.conf`, which is the GDM configuration file, using the nano text editor to allow root login. Add `AllowRoot=true`, as shown in Figure [5-45](#), and save the file.

Task



```
GNU nano 4.8 /etc/gdm3/custom.conf
# GDM configuration storage
#
# See /usr/share/gdm/gdm.schemas for a list of available options.

[daemon]
# Uncomment the line below to force the login screen to use Xorg
#WaylandEnable=false

# Enabling automatic login
# AutomaticLoginEnable = true
# AutomaticLogin = user1

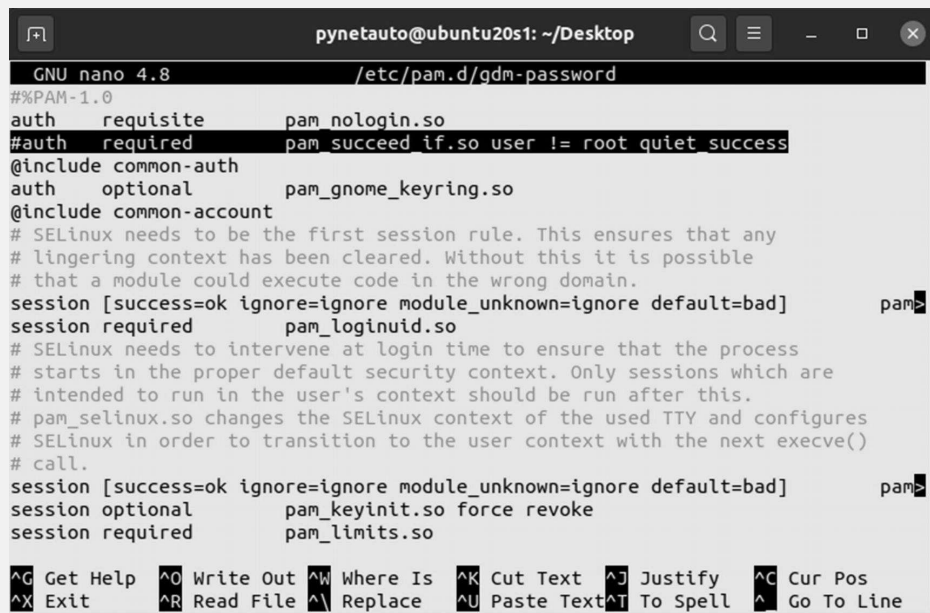
# Enabling timed login
# TimedLoginEnable = true
# TimedLogin = user1
# TimedLoginDelay = 10
AllowRoot=true

[security]
```

Figure 5-45. Ubuntu VM customization 3, updating the gdm3/custom.conf file

- 02 Next, edit the PAM authentication daemon configuration file at `/etc/pam.d/gdm-password` and comment on the specific line that denies root access to the graphical user interface. Use the nano or vi editor and open the `/etc/pam.d/gdm-password` file. Put # at the beginning of the line, which starts with `auth required pam_succeed_if.so user!= root quiet_success`, that you are commenting out, which means you are disabling the feature. See Figure [5-46](#).

Task



```

pynetauto@ubuntu20s1: ~/Desktop
GNU nano 4.8 /etc/pam.d/gdm-password
##PAM-1.0
auth requisite pam_nologin.so
#auth required pam_succeed_if.so user != root quiet success
@include common-auth
auth optional pam_gnome_keyring.so
@include common-account
# SELinux needs to be the first session rule. This ensures that any
# lingering context has been cleared. Without this it is possible
# that a module could execute code in the wrong domain.
session [success=ok ignore=ignore module_unknown=ignore default=bad] pam_
session required pam_loginuid.so
# SELinux needs to intervene at login time to ensure that the process
# starts in the proper default security context. Only sessions which are
# intended to run in the user's context should be run after this.
# pam_selinux.so changes the SELinux context of the used TTY and configures
# SELinux in order to transition to the user context with the next execve()
# call.
session [success=ok ignore=ignore module_unknown=ignore default=bad] pam_
session optional pam_keyinit.so force revoke
session required pam_limits.so

^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
^X Exit ^R Read File ^\ Replace ^U Paste Text ^T To Spell ^_ Go To Line

```

Figure 5-46. Ubuntu VM customization 3, updating the pam.d/gdm-password file

Task

A mandatory reboot is required here once again. Reboot your Ubuntu server using the `sudo reboot` command or using the power button on the top-right corner of your Ubuntu desktop. Once the server has been restarted, log in as another user and enter **root** as the username and enter the root user's password. See Figure [5-47](#).

03



Figure 5-47. Ubuntu VM customization 3, root user GUI login

Task

- 04 Congratulations! You can log into Ubuntu desktop as a root user. See Figure [5-48](#).

Task

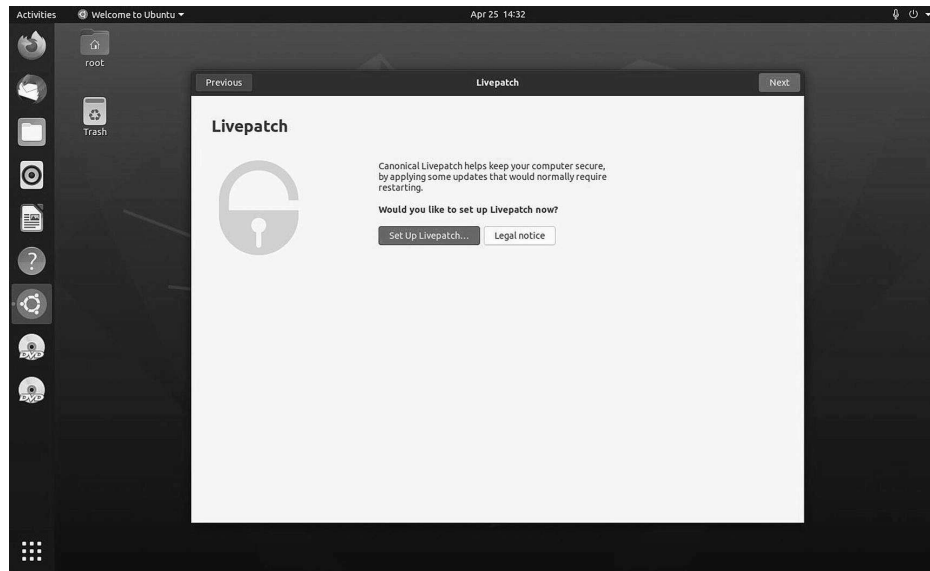


Figure 5-48. Ubuntu VM customization 3, Ubuntu desktop root user login

Taking a Snapshot of a Virtual Machine

One of the most powerful features of running an operating system as a VM is the power of snapshots. In a real production environment with ESXi and vCenter-based solutions, taking snapshots and cloning are potent ways to prevent system failure and recover from failures. In the VMware technology world, other system recovery features include high availability, vMotion, and DRS. But on the VMware Workstation Pro solution, we have snapshots and clones to preserve the current state of our VMs. A snapshot is like taking a photo of your system, so you can reverse unintended or failed system changes back to the prime system state of a VM. The cloning feature allows you to make a partial or full backup of your VMs to preserve the system state in the case of system failure.

For Python network automation labs, you can choose to use a snapshot or full cloning as a fail-safe recovery feature. Snapshots will allow us to move in time to various points during our lab. Follow the next tasks to make a

snapshot of your Ubuntu server before installing other software on the server.

Task

To take a snapshot of your current server state, from your VMware Workstation menu, select VM ► Snapshot ► Take Snapshot. See Figure [5-49](#).

01

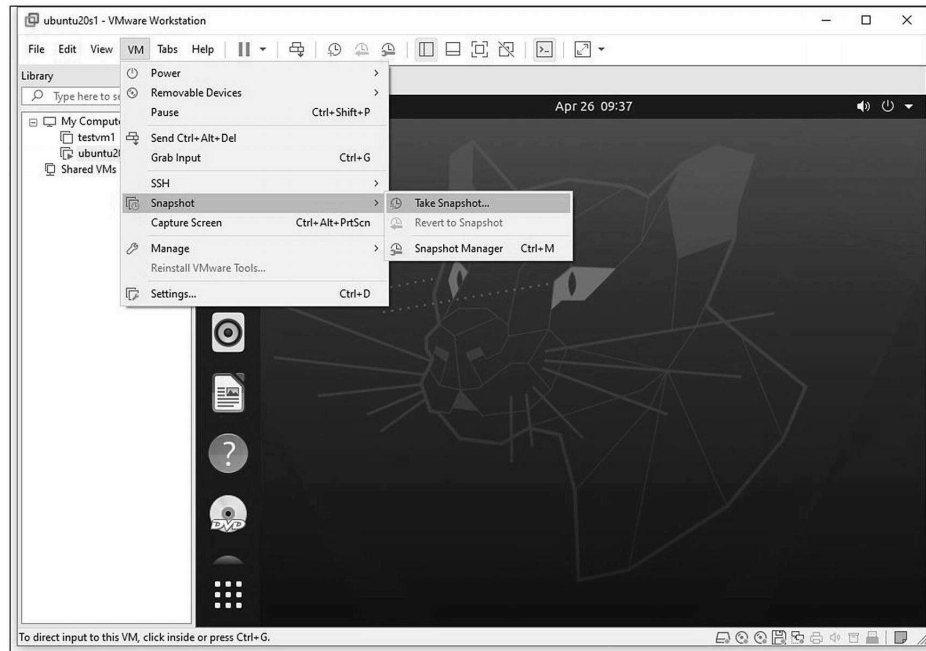


Figure 5-49. VMware snapshot, navigating to the Snapshot menu

02 Enter the name and tasks performed as a description, and click the Take Snapshot button to create a snapshot, as shown in Figure [5-50](#).

Task

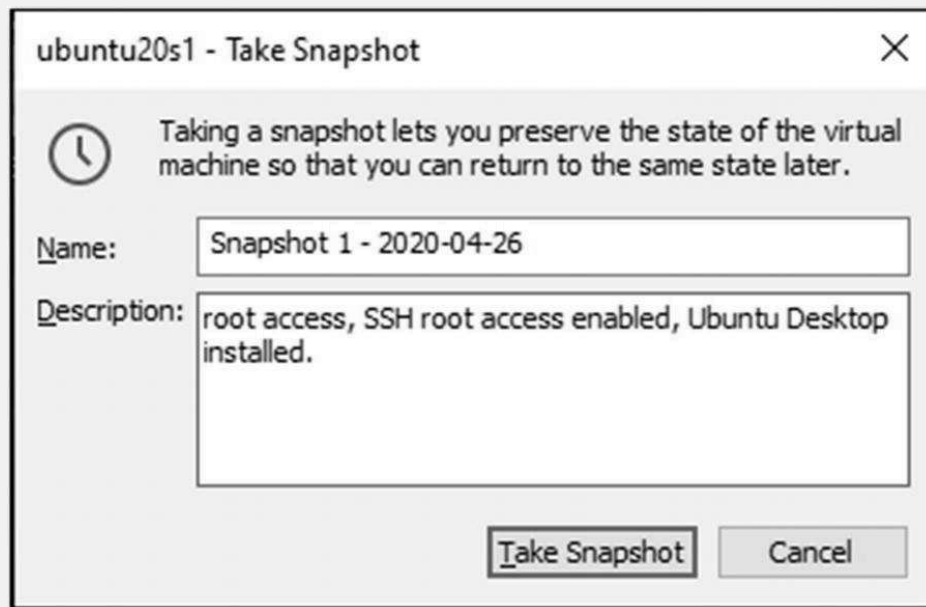
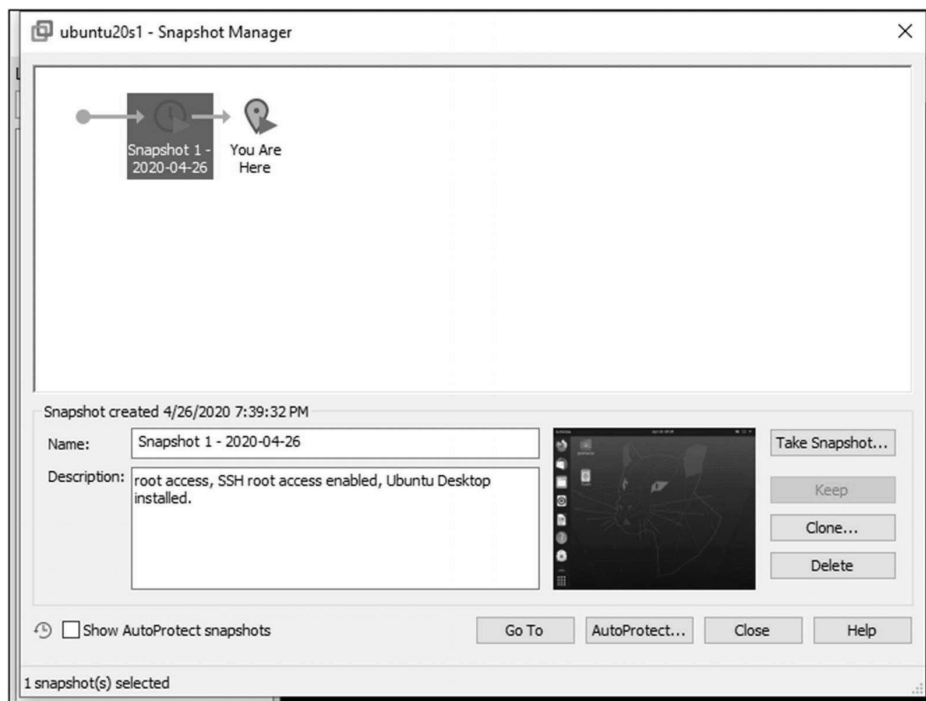


Figure 5-50. VMware snapshot, creating a VM snapshot

- 03 At the bottom-left corner of the window, the percentage of snapshot progress will be displayed. Once it reaches 100 percent, navigate to the Snapshot Manager from the menu, and confirm the snapshot. See Figure [5-51](#).



#	Task
---	------

Figure 5-51. VMware snapshot, confirming snapshot

Cloning a Virtual Machine

Now you have a customized Ubuntu virtual server ; if you want to make another VM, you can do so by cloning the same VM. Cloning allows you to re-create the same machine at the speed of lightning compared to how much time you spent installing a new VM from scratch. Cloning also can make a full backup of the current VM state and preserve the server state. Now let's take a look at this process.

Task

To clone a VM, navigate to VM ► Manage ► Clone. (See Figure 5-52) Alternatively, you can also start the cloning process within the Snapshot Manager.

01

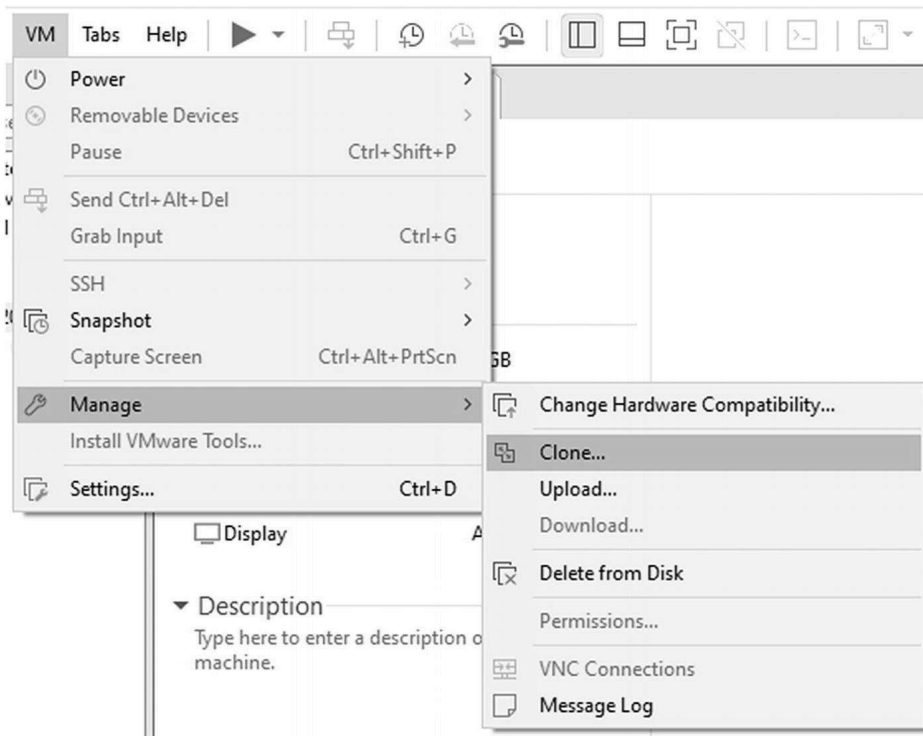


Figure 5-52. VMware cloning 1, making a clone

02 Leave the default settings and click the Next button. See Figure 5-53.

Task

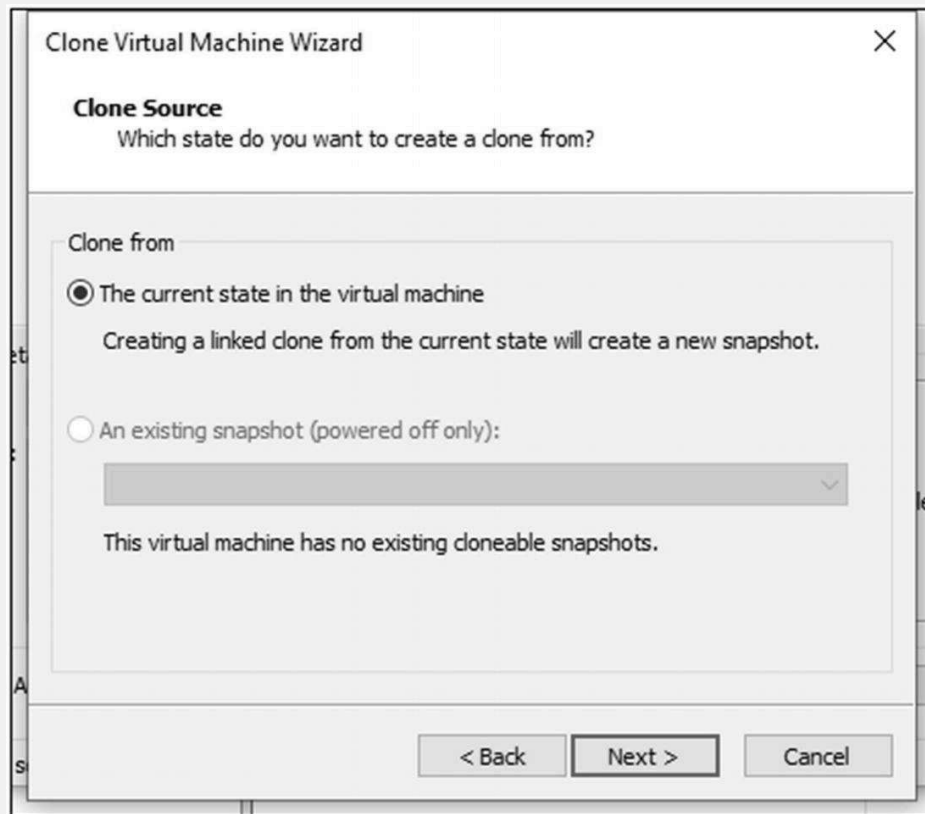


Figure 5-53. VMware cloning 1, clone source

- 03 Select “Create a full clone” and click the Next button. See Figure [5-54](#).

Task

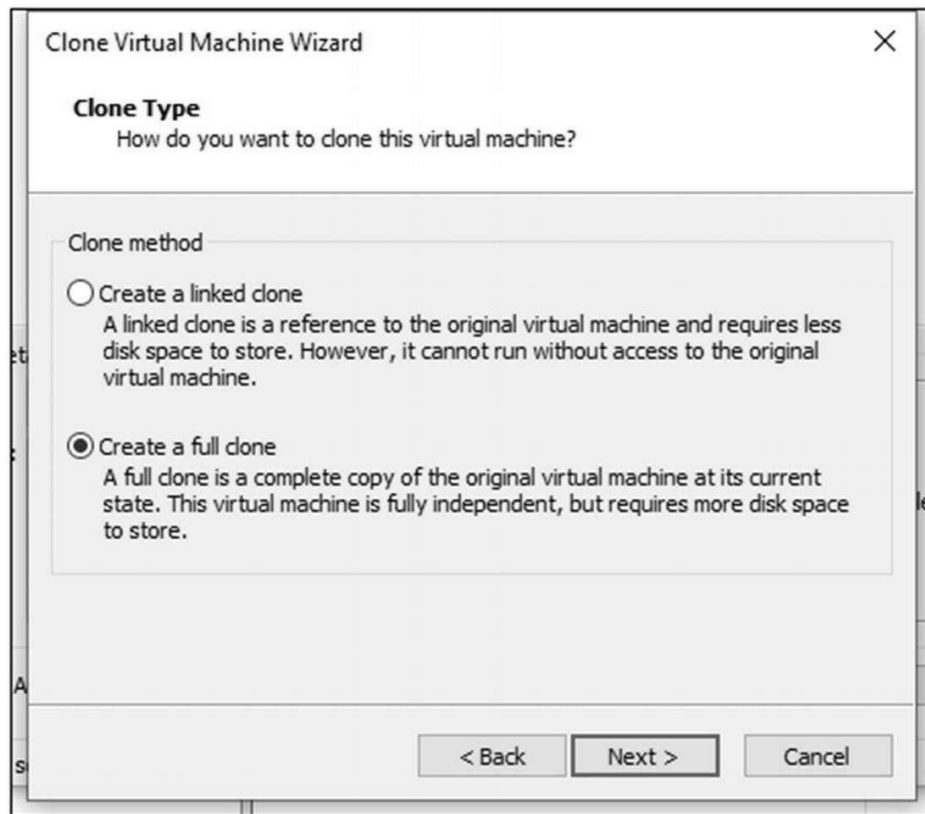


Figure 5-54. VMware cloning 1, clone type, a full clone

04 Rename the cloned VM and click the Finish button. See Figure [5-55](#).

Task

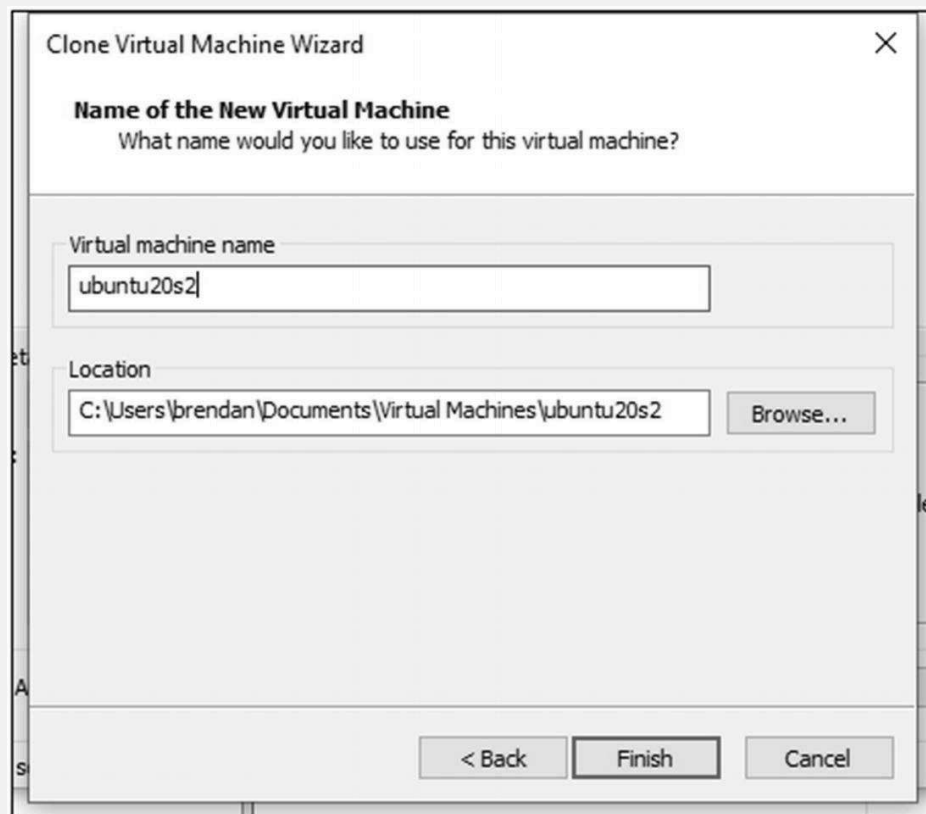


Figure 5-55. VMware cloning 1, renaming the cloned VM

05 Now click the Close button. See Figure [5-56](#).

Task

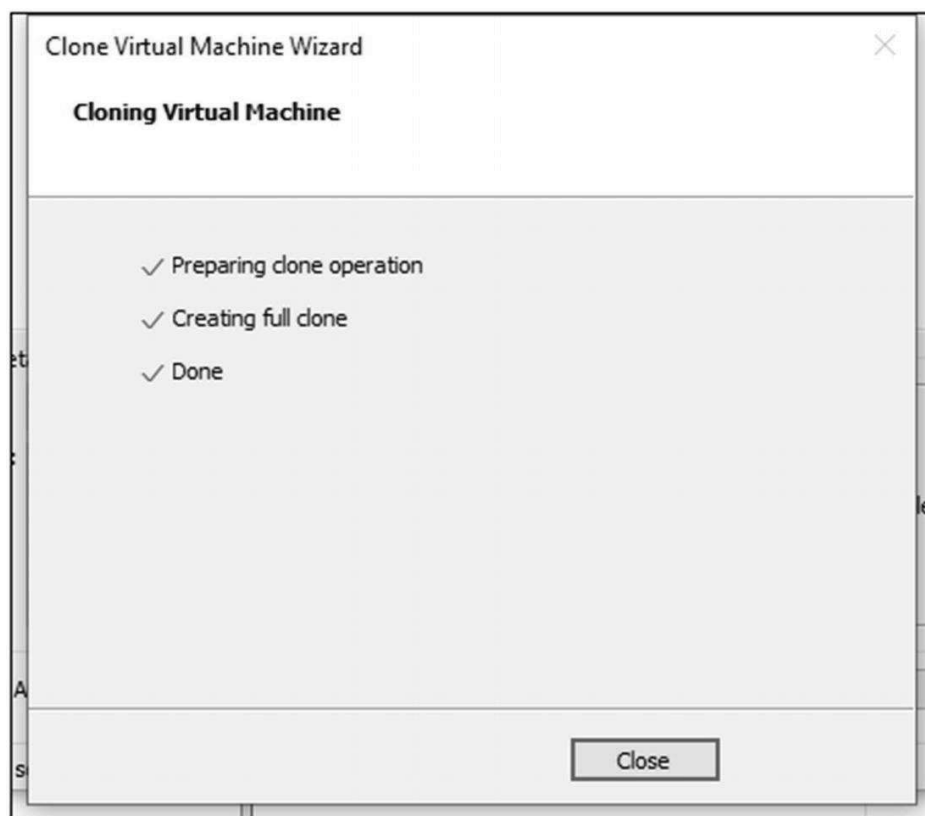


Figure 5-56. VMware cloning 1, closing the VM cloning wizard

- 06 You will see another Ubuntu server added to your VM library. See Figure [5-57](#).

Task

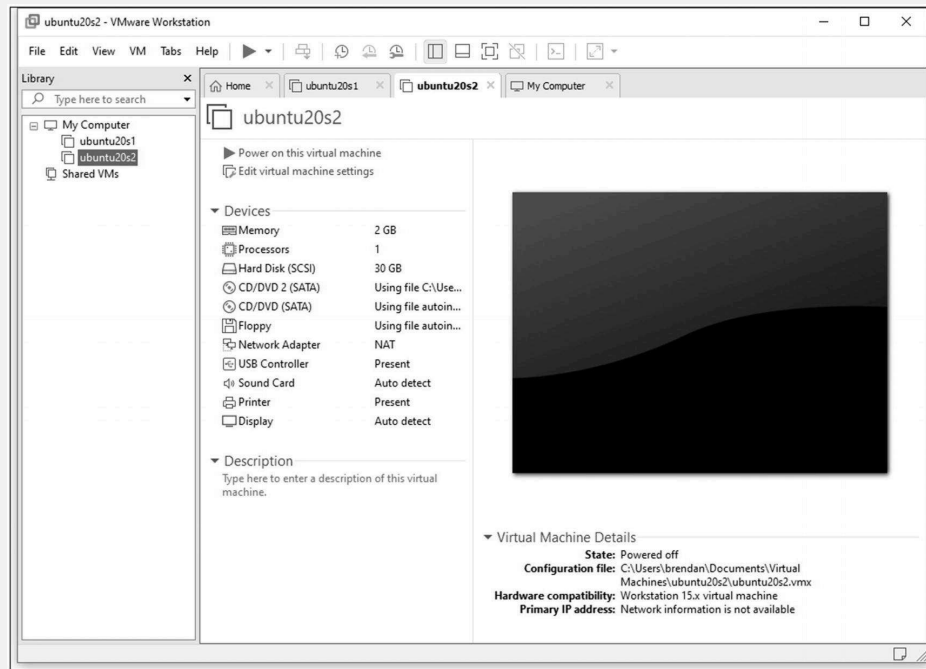


Figure 5-57. VMware cloning 1, cloned VM added to the library

Optionally, if you want to copy and duplicate your VM simply, simply copy the folder containing all the VM files to other storage. This method will take up a lot of hard disk space, so if you are using a smaller SSD, then a snapshot is a better option than fully cloning a VM. See Figure [5-58](#).

07

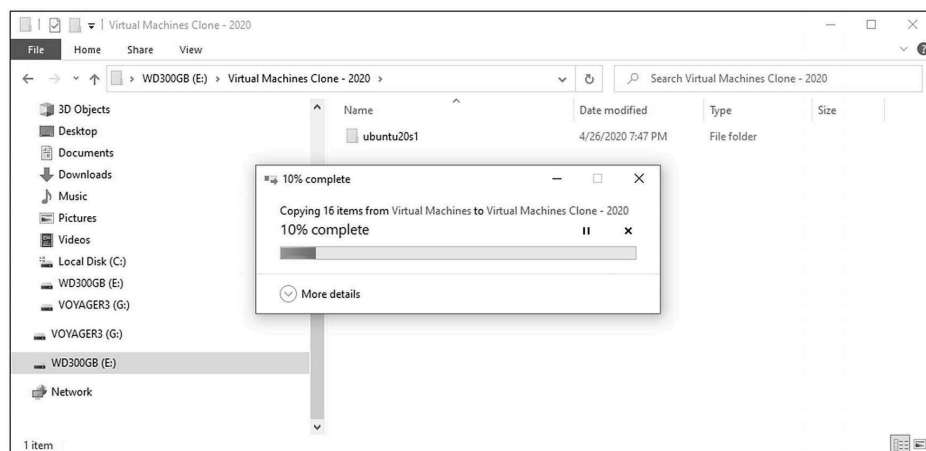


Figure 5-58. VMware cloning 2, copying the VM folder to an external drive

Summary

Installing an Ubuntu Server as a virtual machine is a straightforward process, but installing a graphical user interface is an option. Now you have successfully installed, customized, backed up, and cloned a newly created Ubuntu server VM, and it is time to create another Linux server as our second server. In the next chapter, you will follow similar steps to create a virtual machine for CentOS 8.1, which will serve as an all-in-one IP services server for our network automation labs.

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